

Glacial hazards

Risk assessment and mitigation

J. F. (Jakob) Steiner

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fidelsteiner.github.io/#teaching

Topics

- 1) Terminology: Hazards. Risks. Mitigation.
- 2) *Global ice and exposure to hazards*
- 3) *Types of glacial hazards*
- 4) *Monitoring and risk mitigation*
- 5) *Three focus excursions:*
 - a) *Mental health and glacial risks (Pakistan)*
 - b) *The politics of national risk management in high mountains (Nepal)*
 - c) *The global politics of 'glacial risks' (Greenland)*

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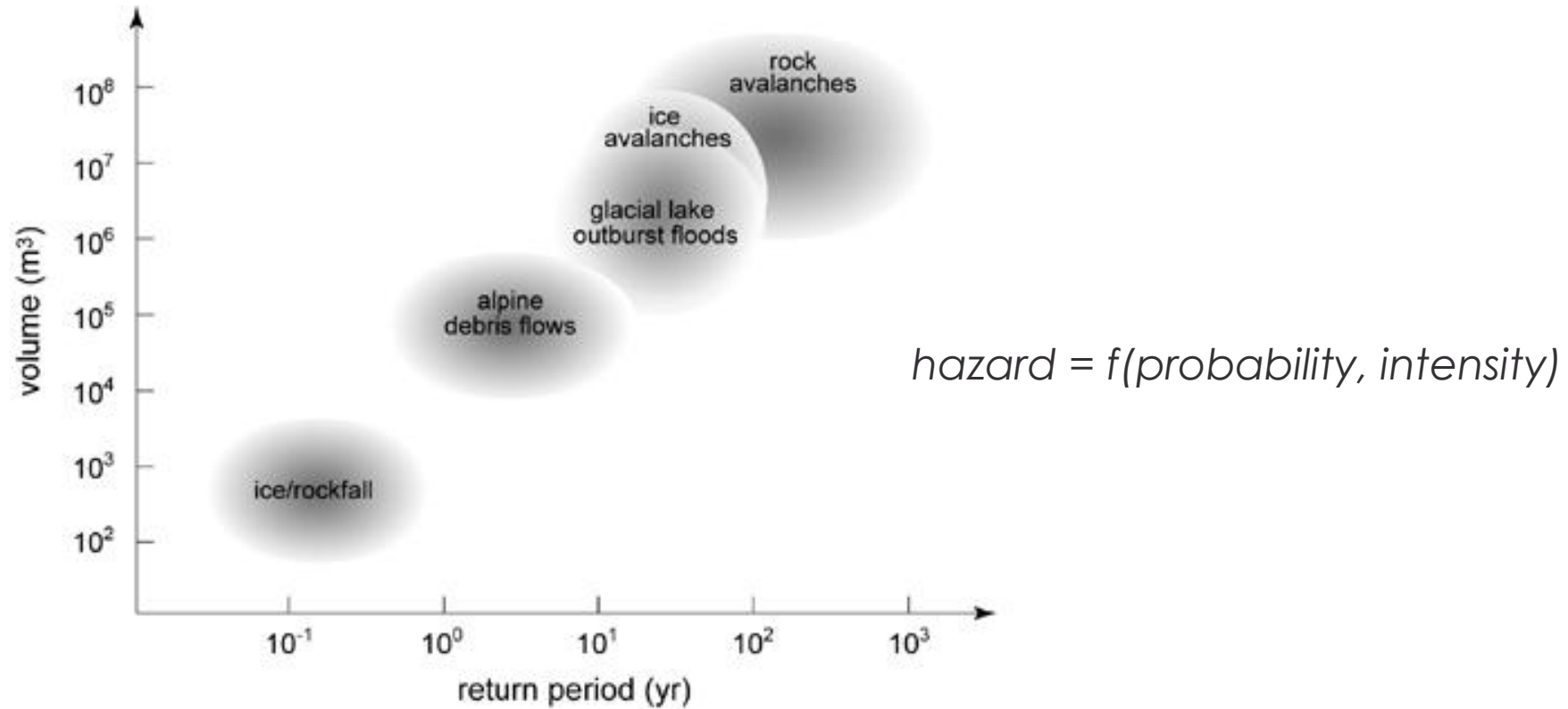
Glacial hazards vs cryosphere hazards

Cryosphere = glaciers – snow – permafrost – sea/lake ice

Other cryosphere hazards are...

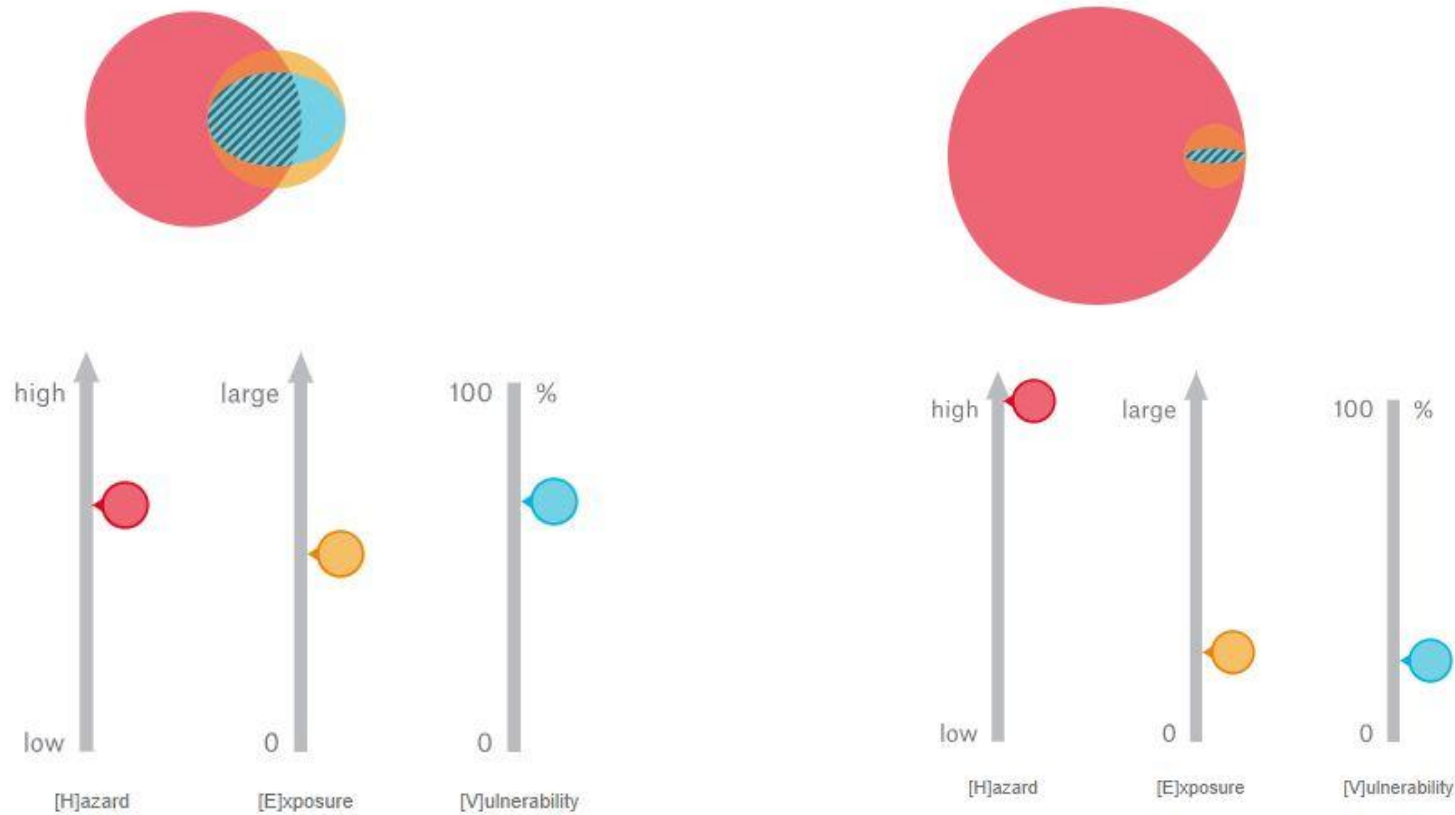
- snow avalanches
- snow droughts
- permafrost change induced slope instabilities
- change of sea ice breakup

Return periods vs. volumes



Geology Today, Volume: 28, Issue: 3, Pages: 98-104, First published: 25 May 2012, DOI: (10.1111/j.1365-2451.2012.00836.x)

risk = ***hazard*** * ***exposure*** * ***vulnerability***
[****adaptive capacity*** ****insurance penetration***]



Mitigation and adaptation

Mitigation – reducing risk from the occurrence of any undesirable event or minimize the degree of damage

Adaptation – prepare for damages that are to be expected (and possibly can't be avoided or mitigated)

Topics

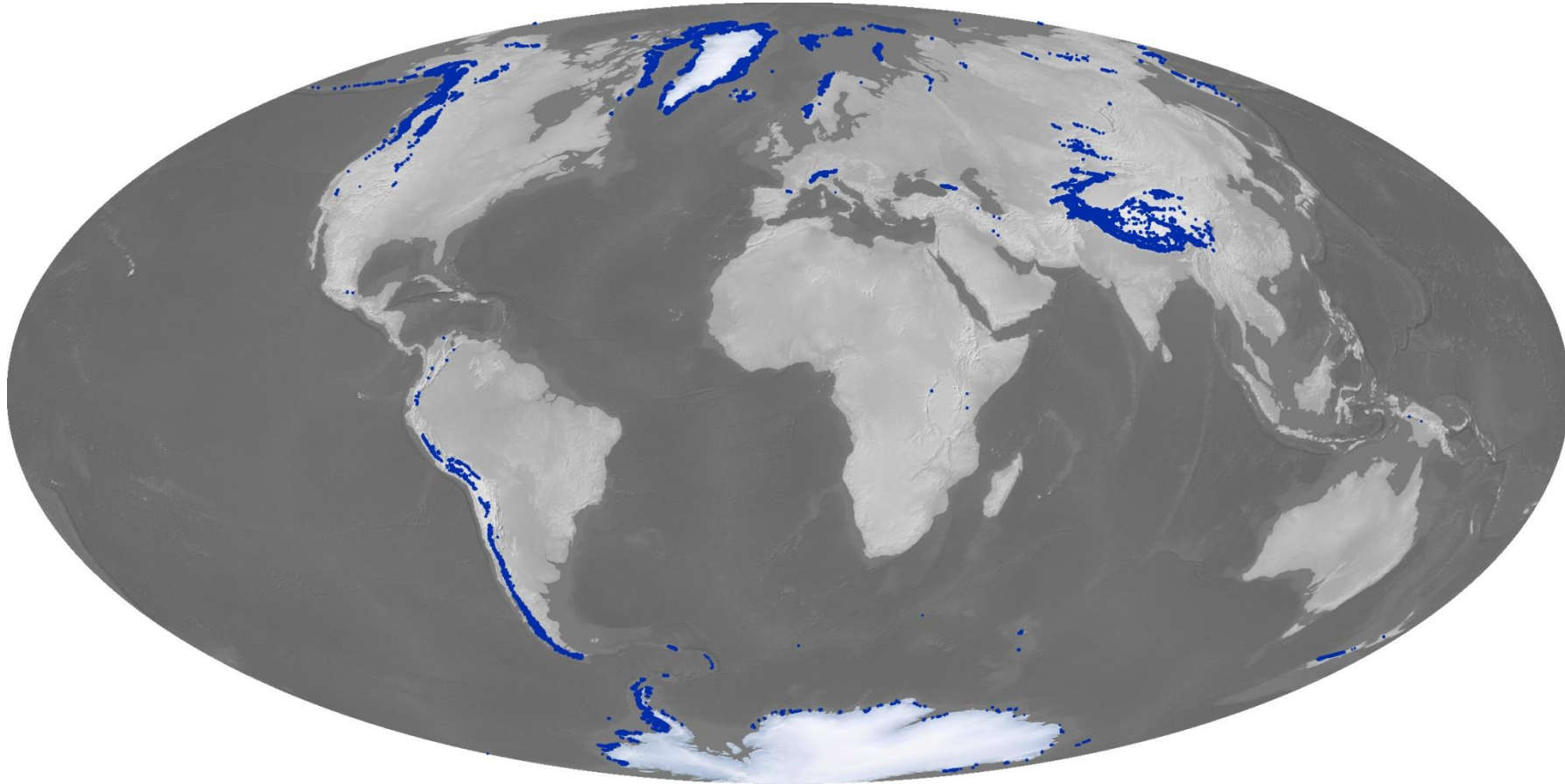
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Glaciers globally

Total Area:

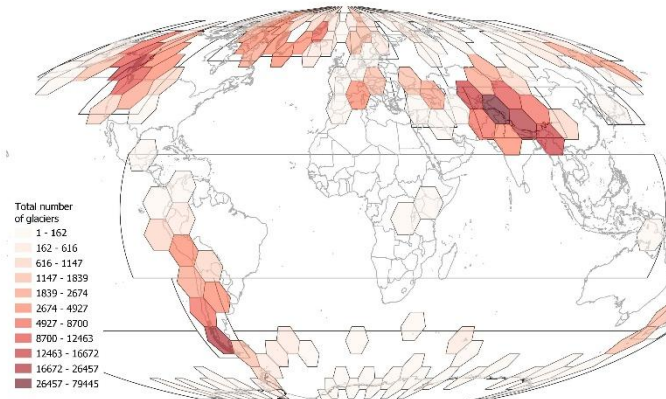
RGI: 734 933 km²

WGI: 747 688 km²

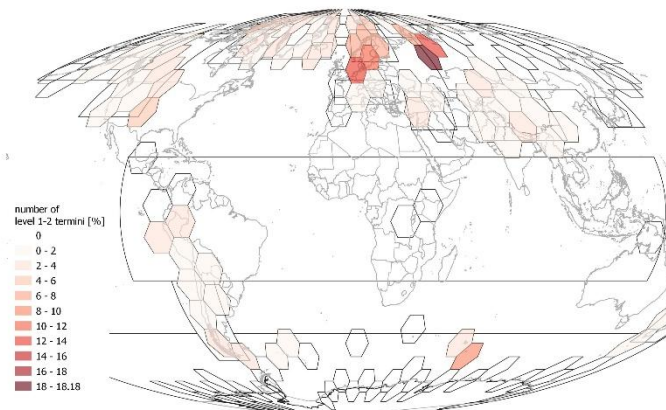


R. Simmon, NASA; based on RGI4
(2013)

Glaciers ending in lakes globally



A - total glacier number



C - relative number lake-terminating glaciers

Global mapping of lake-terminating glaciers

Jakob Steiner^{1,2}, William Armstrong^{3,*}, Will Kochitzky^{4,*}, Robert McNabb^{5,*}, Rodrigo Aguayo⁶, Tobias Bolch⁷, Fabien Maussion⁸, Vibhor Agarwal^{9,10}, Iestyn Barr¹⁰, Nathaniel R. Baurley¹¹, Mike Cloutier¹², Katelyn DeWater⁴, Frank Donachie¹³, Yoann Drocourt¹⁴, Siddhi Garg¹⁵, Gunjan Joshi¹⁶, Byron Guzman¹⁷, Stanislav Kutuzov^{18,19}, Thomas Loriaux²⁰, Caleb Mathias²¹, Brian Menounos²¹, Evan Miles^{22,23,24}, Aleksandra Osika²⁵, Kaleigh Potter⁴, Adina Racoviteanu²⁶, Brianna Rick²⁷, Miles Sterner¹², Guy D. Tallentire²⁸, Levan Tielidze^{29,30}, Rebecca White³¹, Kunpeng Wu^{7,32}, and Whyjay Zheng³³

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²Asian Mountain Academic Alliance, Lalitpur, Nepal

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⁴School of Marine and Environmental Programs, University of New England, Biddeford, Maine, USA

⁵School of Geography and Environmental Sciences, Ulster University, Coleraine, UK

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⁷Institute of Geodesy, Graz University of Technology, Graz, Austria

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¹⁷Universidad Austral de Chile, Valdivia, Chile

¹⁸School of Earth Sciences, Ohio State University, Columbus, USA

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²¹Department of Geography, Earth, and Environmental Sciences, University of Northern British Columbia, Prince George, British Columbia, Canada

²²Department of Geography, University of Zurich, Zurich, Switzerland

²³Department of Geosciences, University of Fribourg, Fribourg, Switzerland

²⁴Swiss Federal Institute for Forest, Snow, and Landscape Research, Birmensdorf, Switzerland

²⁵Institute of Earth Sciences, University of Silesia in Katowice, Katowice, Poland

²⁶Université Grenoble Alpes, CNRS, IRD, IGE – Saint Martin d'Hères, Saint Martin d'Hères, France

²⁷RedCastle Resources, Salt Lake City, Utah, USA

²⁸Department of Geography and Environment, Loughborough University, Loughborough, UK

²⁹Securing Antarctica's Environmental Future, School of Earth, Atmosphere and Environment, Monash University, Clayton, Australia

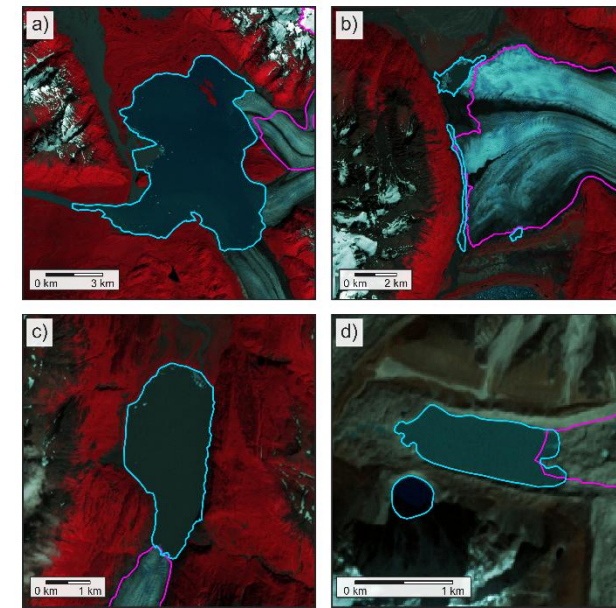
³⁰School of Natural Sciences and Medicine, Ilia State University, Tbilisi, Georgia

³¹School of Geography, University of Leeds, Leeds, UK

³²Institute of International Rivers and Eco-Security, Yunnan University, Kunming, China

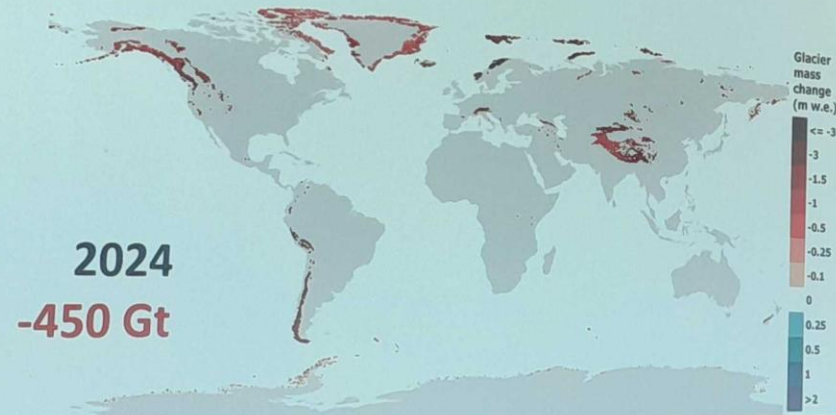
³³Center for Space and Remote Sensing Research, National Central University, Taoyuan, Taiwan

*These authors contributed equally to this work.



Steiner et al. 2026

How fast are glaciers melting?

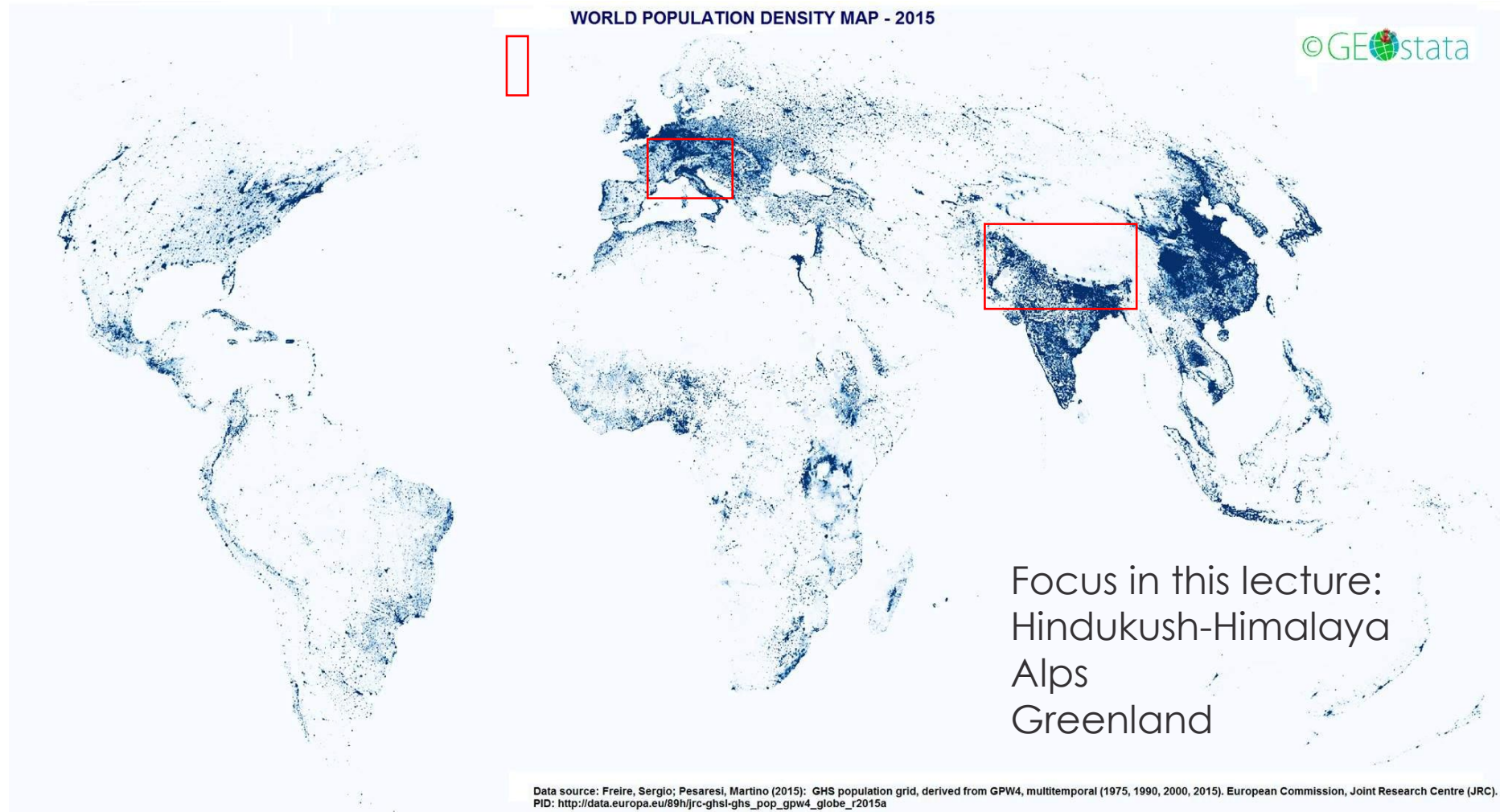


We can by now document
change rapidly

1928's initiative



... but it also pays to
think ahead



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Glacier related hazards

Glacier related hazards

glacial lake outburst floods (+ resulting events)

ice break offs (+ resulting events) / collapsing glaciers

glacier length variations

destabilization of frozen soils/slopes

... what appears when the ice is gone?

Kaab et al., 2005

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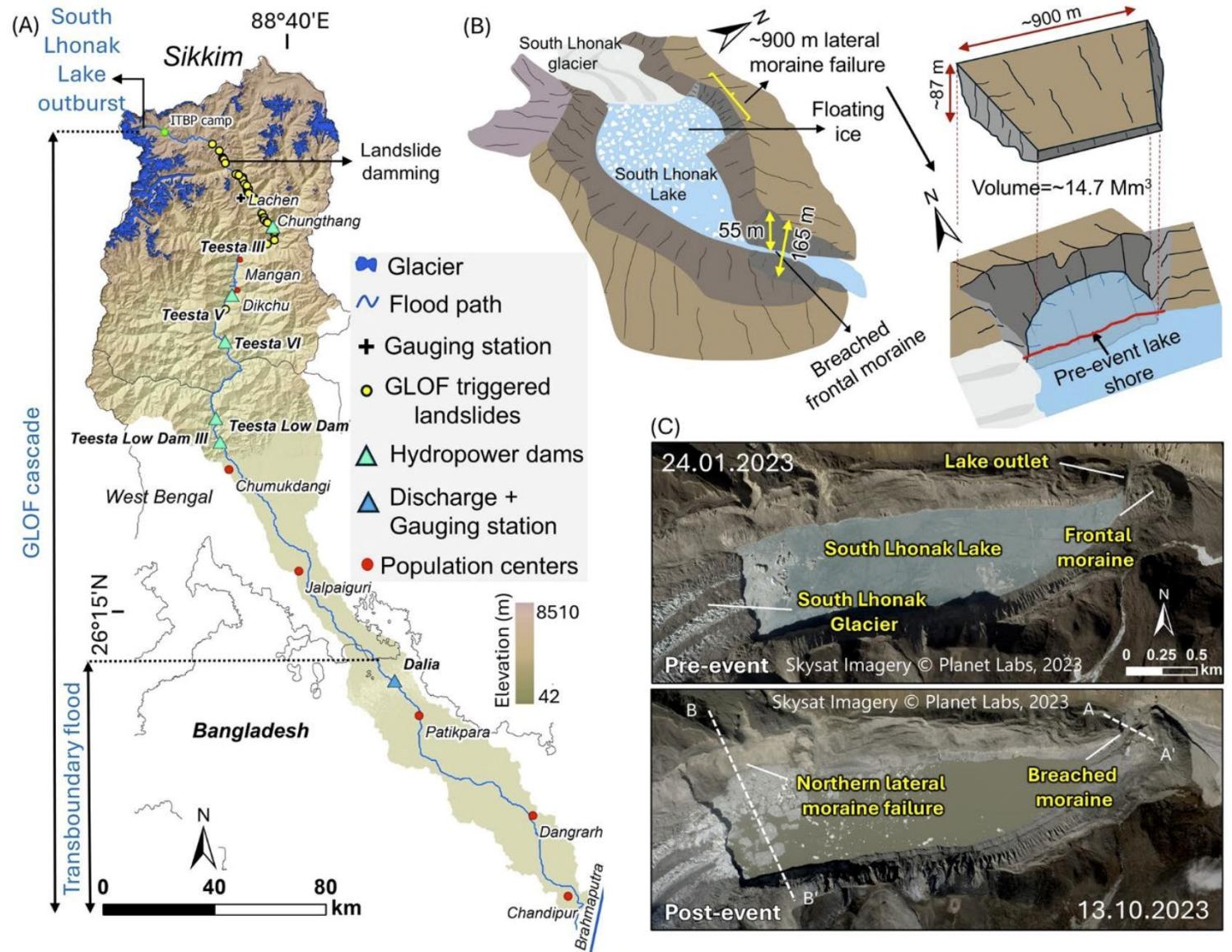
Kaab et al., 2005

Glacial Lake Outburst Floods

British Columbia
Canada

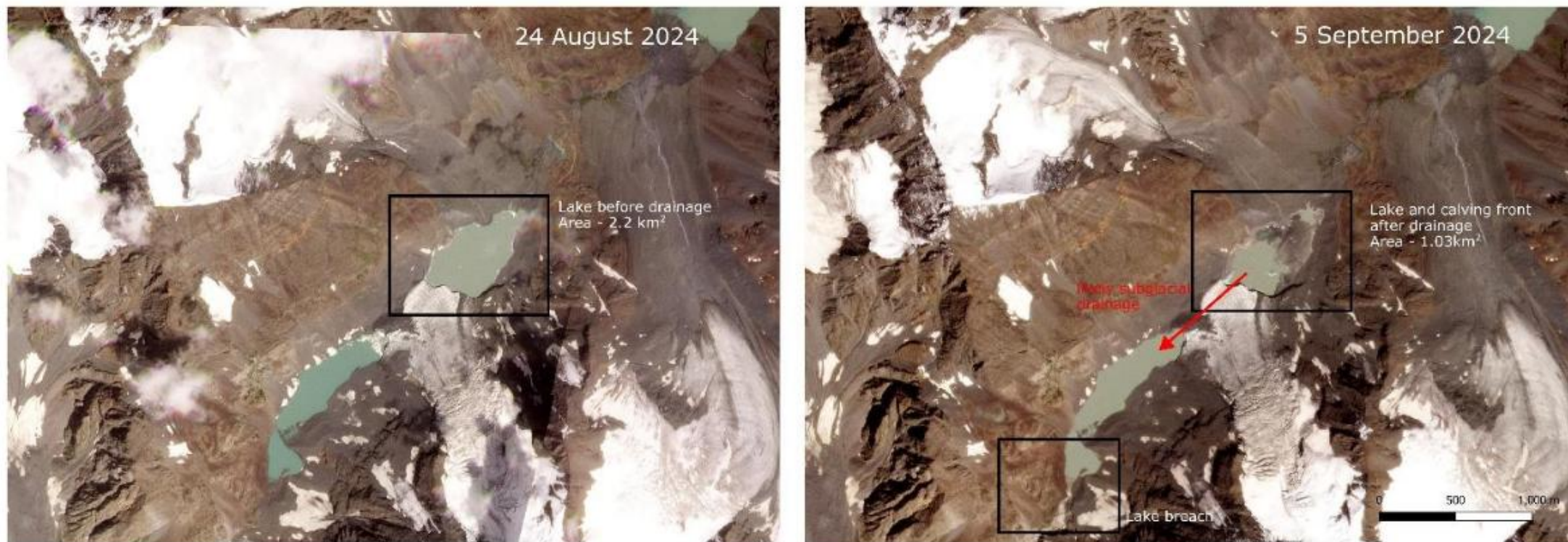


Fig. 12. Aerial photographs of Nostetuko Lake before (left, BC79069-190) and after (right, BC86048-147) the outburst flood of July 19, 1983. Failure occurred when



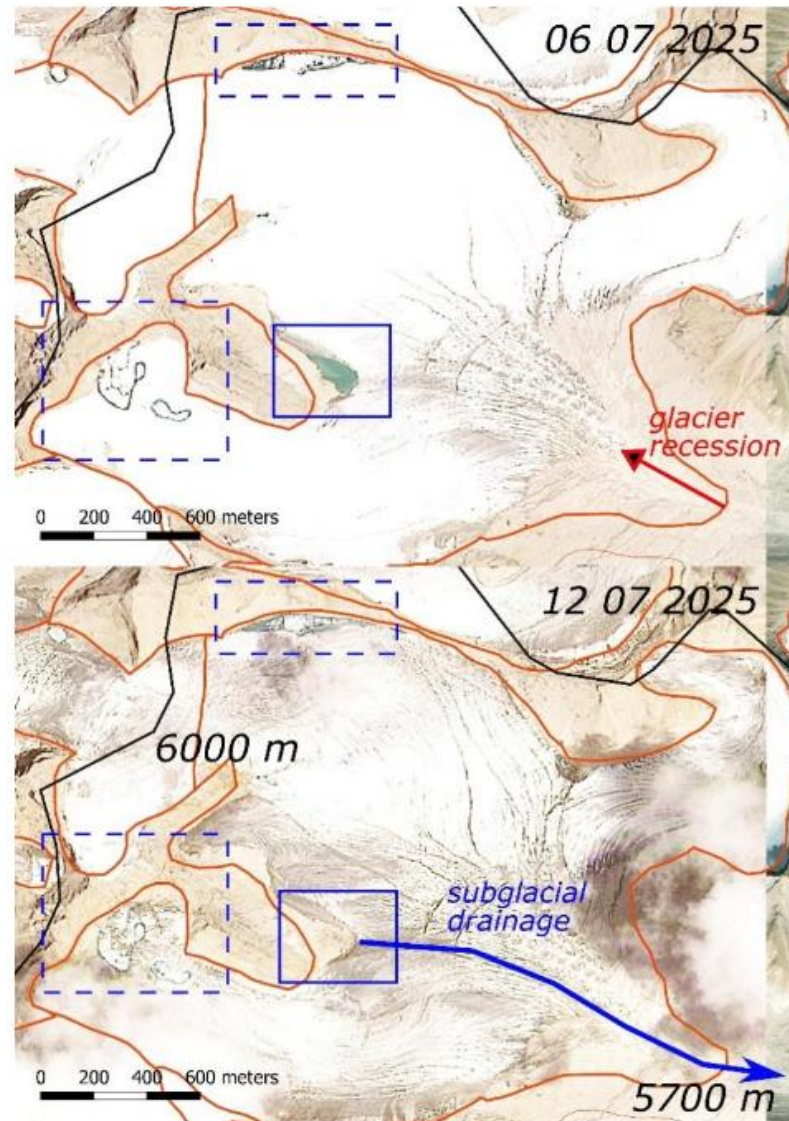
Sattar et al. 2025;

انځورونه: ذریعہ، Planet SuperDove: (پلېنېټ سپر ډوب "یوه سیاره") 25th of August, 4th and 5th of September (پنځویشتم اگست کال دوه زره څلیر شتم)



نمونه "انځور": 1 ډنډ د شلېدلو نه مخکېني او وروستو .

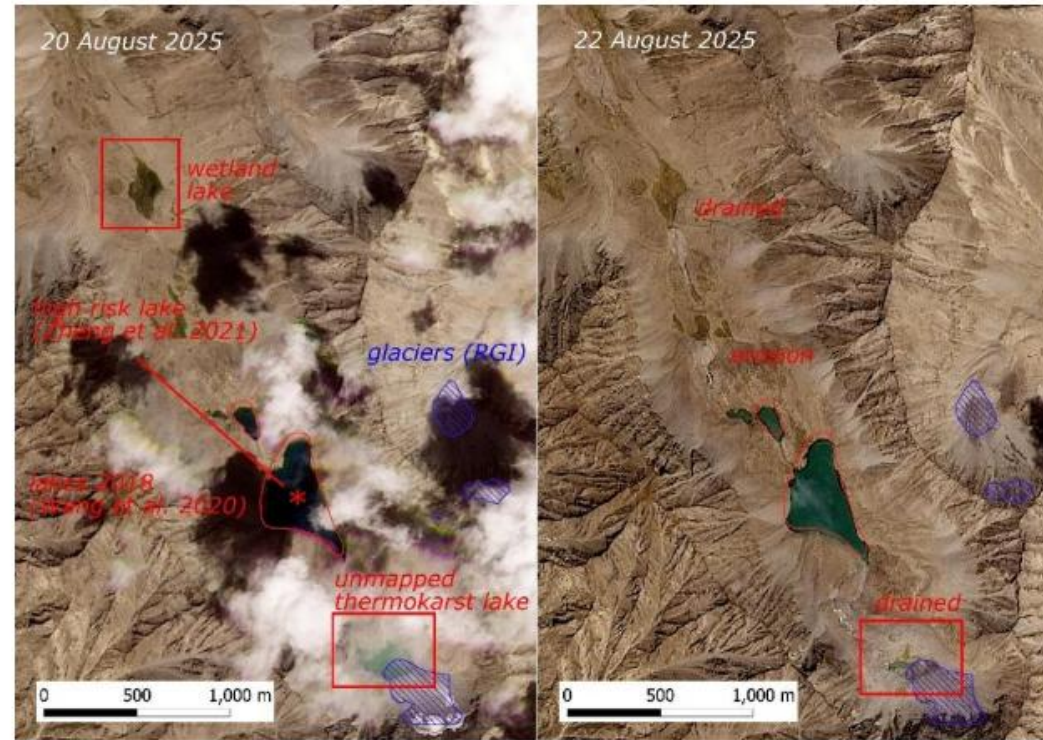
https://hirisk.org/wp-content/uploads/2024/11/RHA_PK3_KumratGLOF-1.pdf



चित्र २: ताल विस्फोट हुनु अघि ६ जुलाईमा देखिएको हिमनदी र प्रष्ट रूपमा देखिने पानी निकासी हुने कुलोहरू (माथिल्लो) र विस्फोट पछि १२ जुलाईमा देखिएको ताल (तल्लो)। निलो थोप्ले बाकसले अझै निकासी नभएका तालहरू देखाउँछ। गाढा रातो रंगको रेखाले सन् २००९ देखि २०२५ सम्म हिउँ बरफ पग्लिएर माथि माथि सड्दै गएको हिमनदीलाई अंकित गरेको छ।

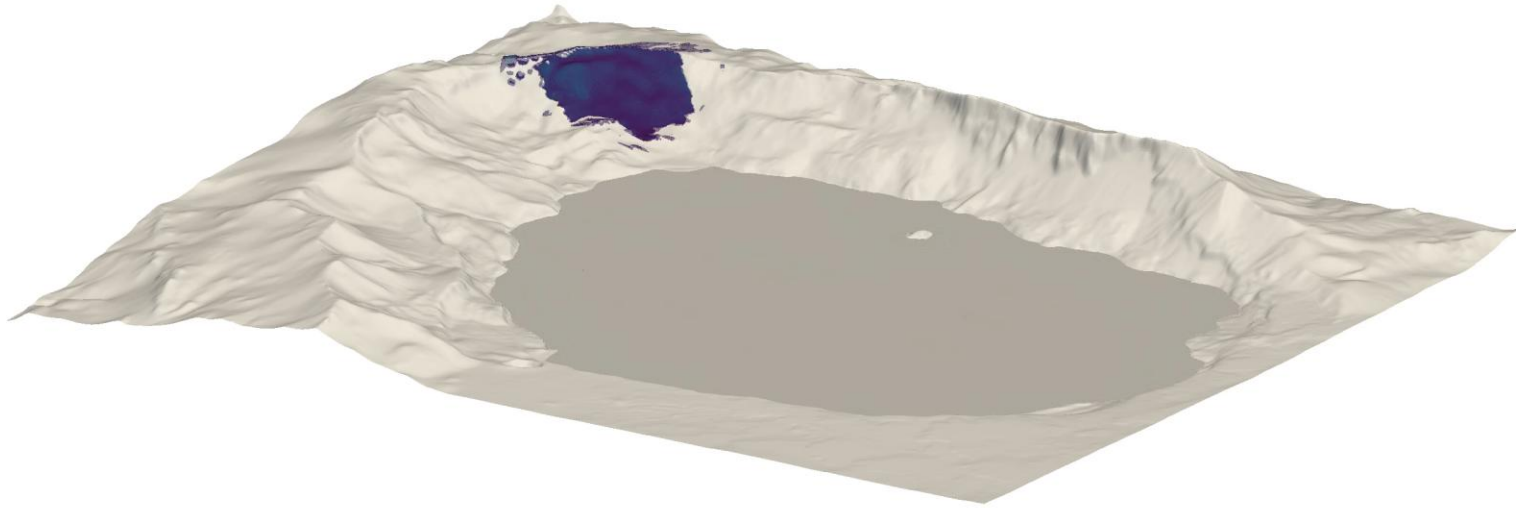


زیادہ رہتا ہے اور اس کے نتیجے میں برف کا احاطہ کم ہوتا ہے (شکل 4)، جس کے نتیجے میں گلیشیئر پگھلنے کی شرح بڑھ جاتی ہے، اور یہی وجہ جھیل کے بہنے کے امکان میں اضافہ کرتی ہے۔



شکل 2: ایونٹ سے پہلے (بائیں) اور بعد (دائیں) میں علاقہ کا منظر۔

Impact waves



Rauter et al., 2022

Glacial Lake Outburst Floods - Causes

Mass movement into the terminal lake causing overtopping (destabilizing glacier/rock slopes)

Failure of a moraine dam (melting of ice cored moraine)

Surging glaciers damming a river (surges/melt season)

Failure of an englacial lake (internal melt)

Intense melt/rainfall

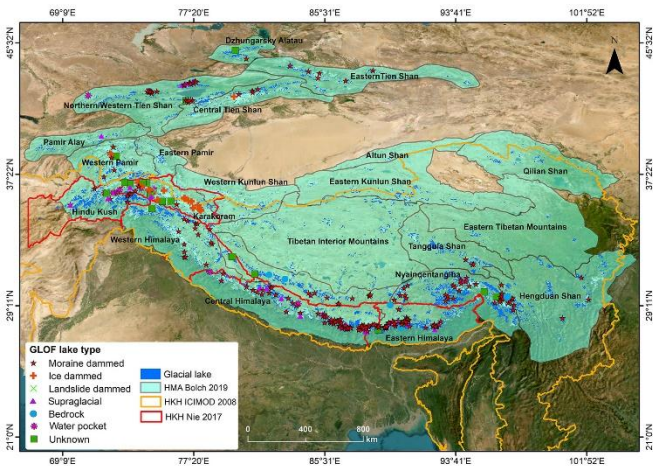
Glacial Lake Outburst Floods - Causes

In High Mountain Asia ca 1000 people died in the last 100 years

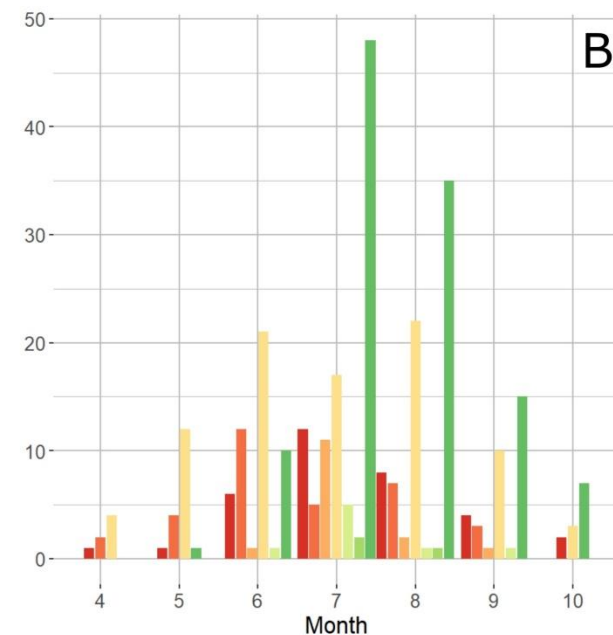
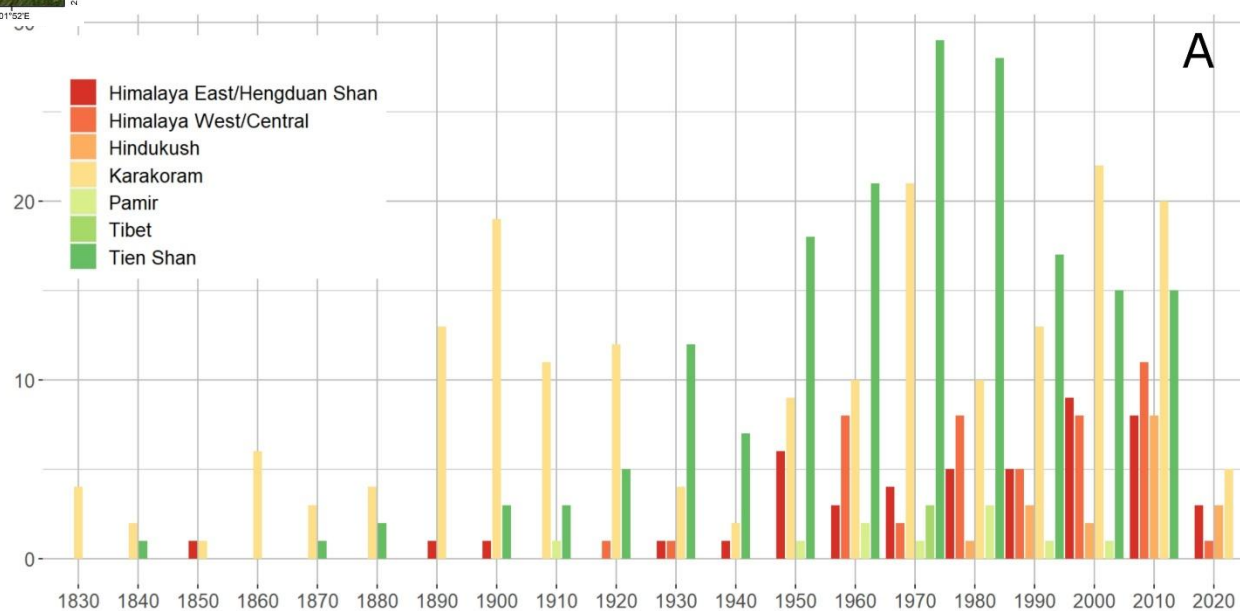
Low frequency – high impact events

(Avalanches: 3000 people died, high frequency, medium impact;

Debris flows: preliminary 8000 people died, high frequency, varied impact)

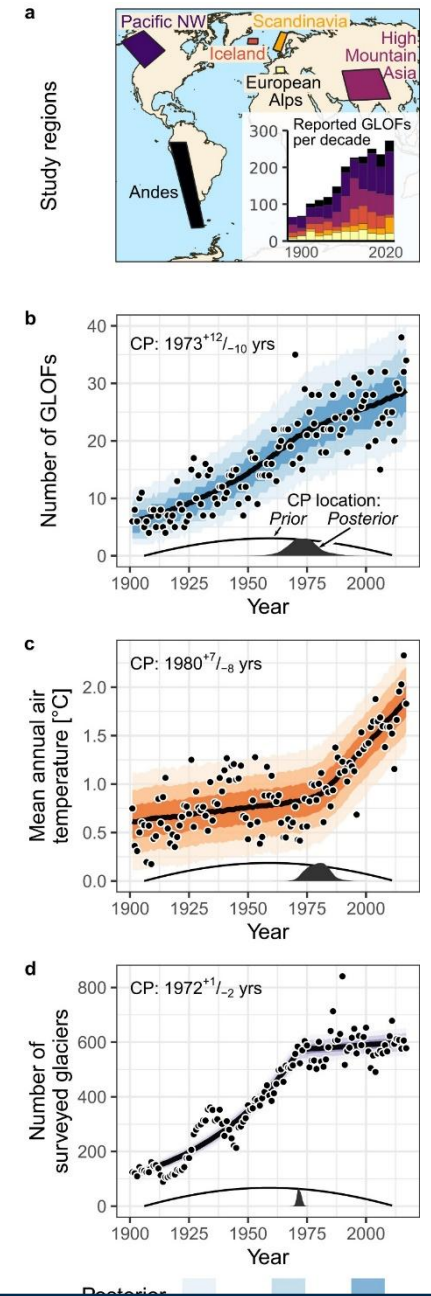


Shrestha et al., 2023



Glacial Lake Outburst Floods - Causes

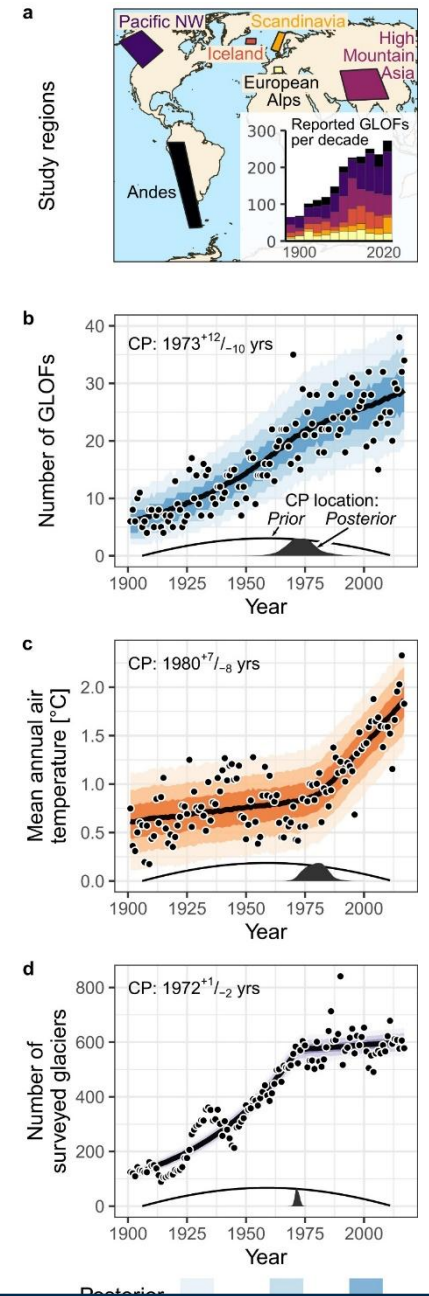
- Link to climate change?



Veh et al., 2022

Glacial Lake Outburst Floods - Causes

- Link to climate change?
 - Seems very plausible and some indications
 - Difficult to proof with very little data
 - Data bias towards more observations in recent years



Veh et al., 2022

Glacial Lake Outburst Floods - *Results*

Lake outbursts from moraine dammed lakes are stagnating, possibly increasing (Veh et al., 2022)

Lake outbursts from ice dammed lakes are decreasing (Veh et al., 2023)

Lake outbursts are decreasing in size, although lakes are increasing in size (Veh et al. 2025)

Outbursts from inside the ice (*waterpockets*) are an uncertain thing we need to investigate further

Glac

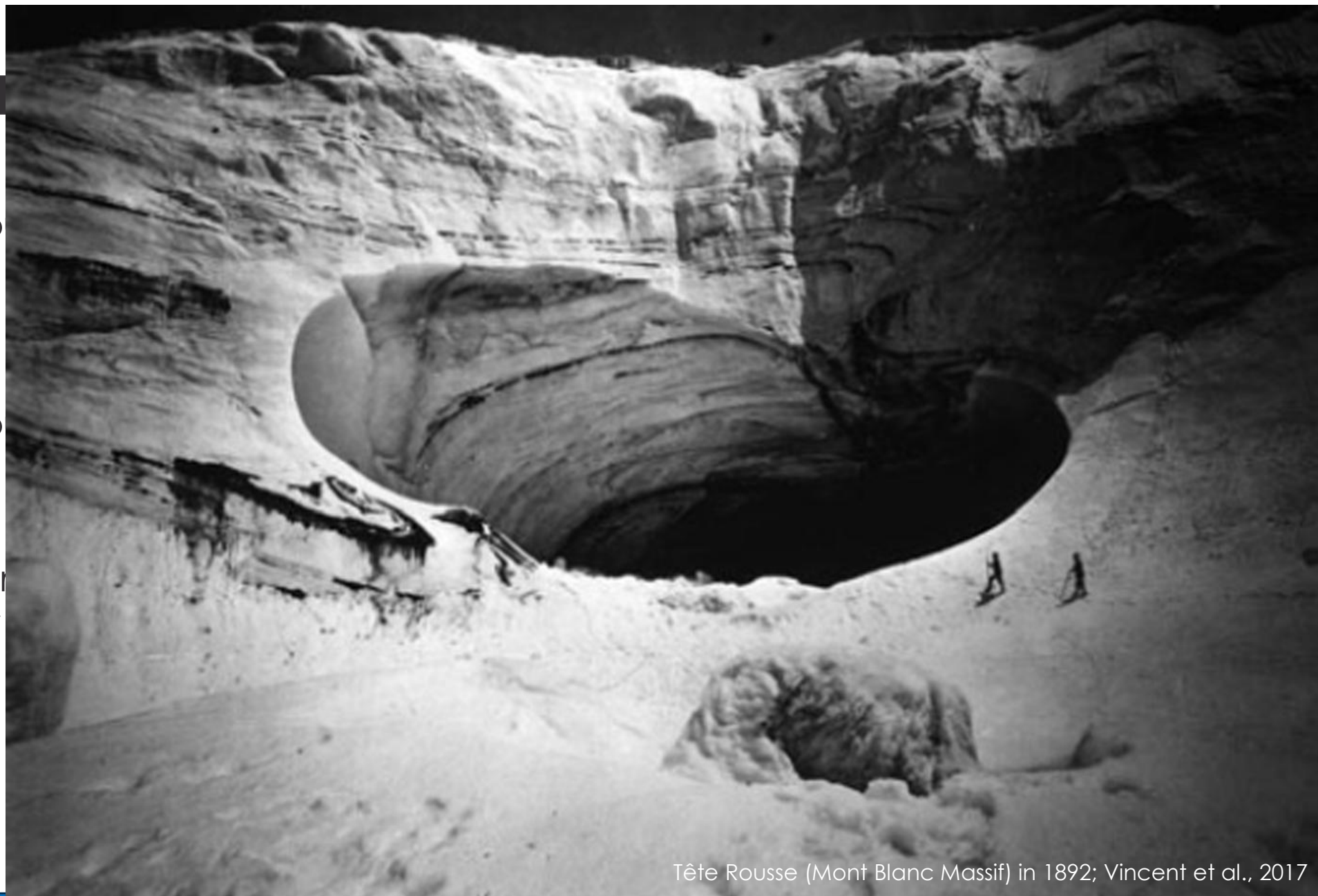
Lake o
2022)

et al.,

Lake o

Outbur
further

igate



Tête Rousse (Mont Blanc Massif) in 1892; Vincent et al., 2017

Glacier related hazards

glacial lake outburst floods (+ resulting events)

ice break offs (+ resulting events) / collapsing glaciers

glacier length variations

destabilization of frozen soils/slopes

... what appears when the ice is gone?

Kaab et al., 2005

Ice break offs + collapsing glaciers

- Increasingly dangerous with receding glacierettes/hanging glaciers
- Added danger in seismically active regions

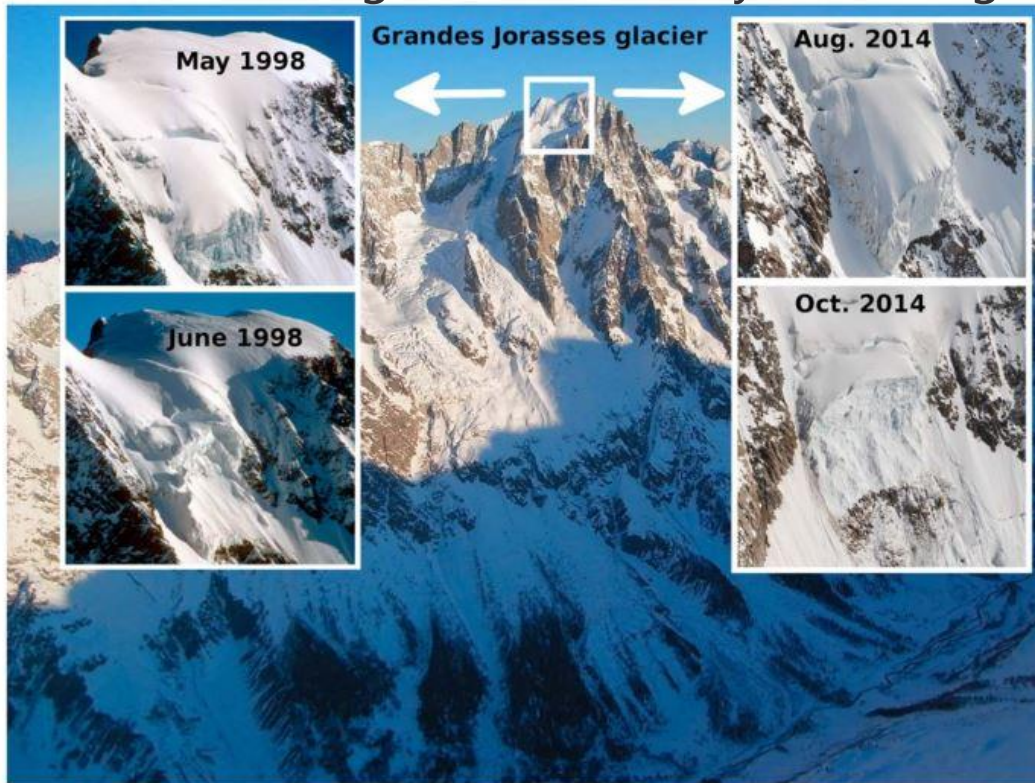


Figure 7. South side of the Grandes Jorasses and the Italian Val Ferret. (left inset) Evolution of the hanging glacier from May to June 1998. (right inset) Evolution of the hanging glacier from August to October 2014.

Faillottaz et al., 2015

Ice break offs + collapsing glaciers





Terminology

Global ice

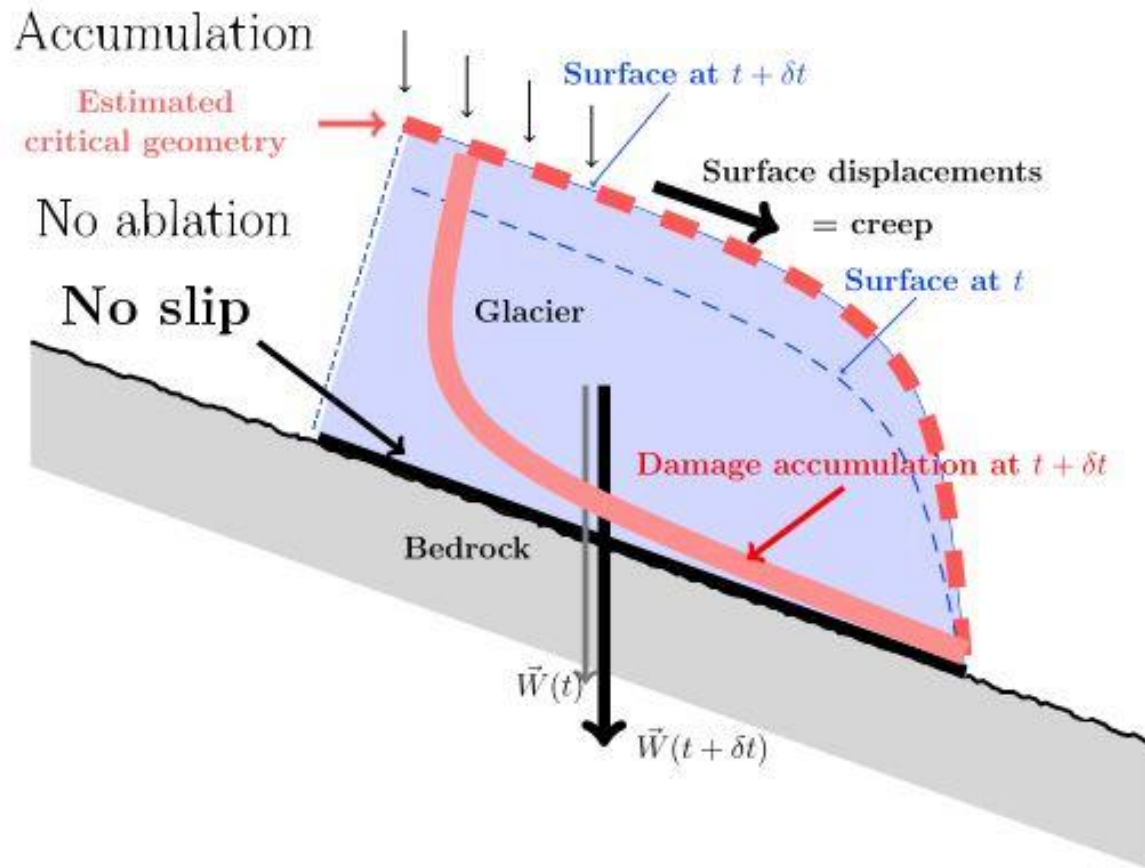
Glacial hazards

Risk management

Case studies

Causes – *cold based; damage propagation*

- 3 glacier types
 - cold based
 - warm based
 - polythermal



Faillottaz et al., 2015

Causes – *transition from cold to warm based*

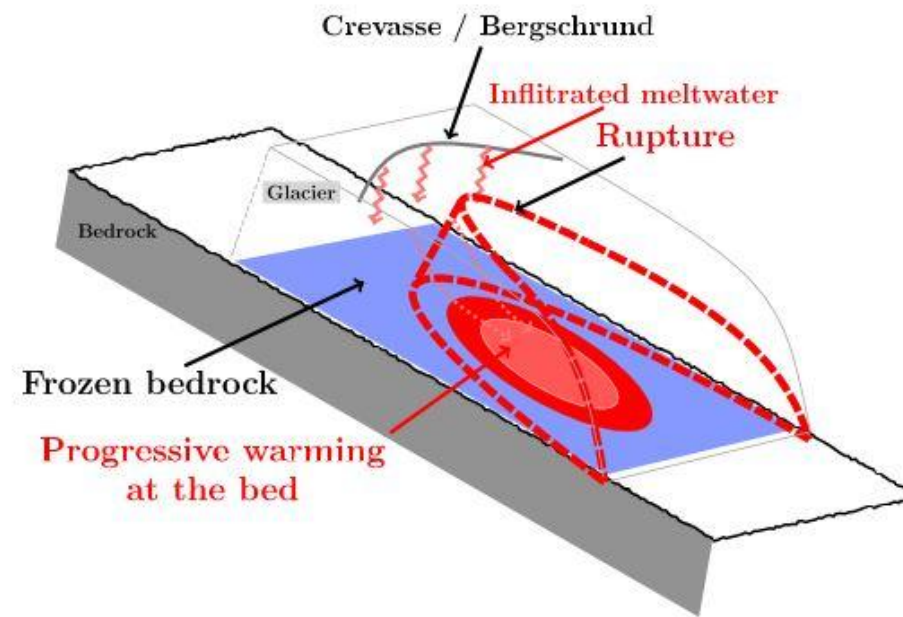


Figure 12. Schematic evolution of the instability initiated by a rapid localized warming at the ice-bedrock interface.

Faillottaz et al., 2015

Causes – *warm based; hydraulic slip*

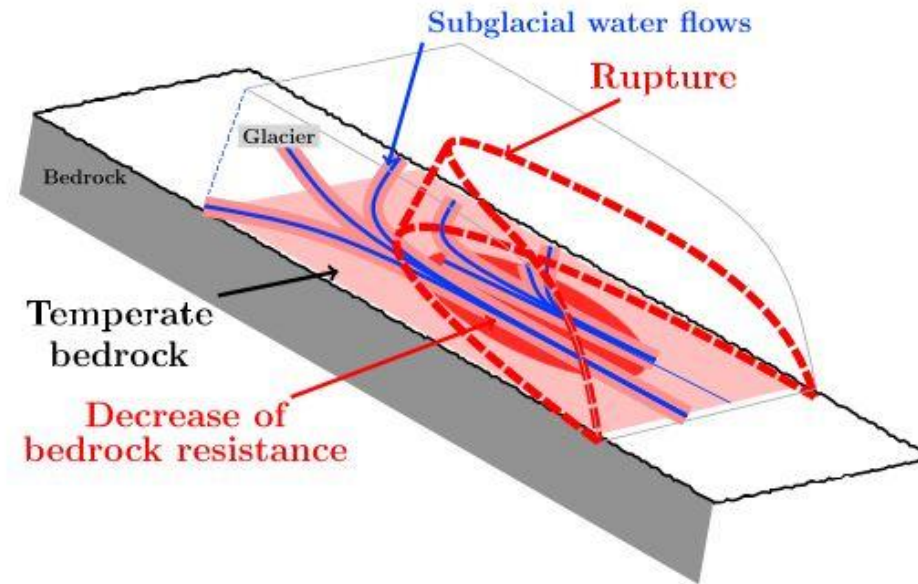


Figure 16. Schematic evolution of the instability for the case of a balanced glacier with a decreasing basal resistance due to increasing subglacial water pressure.

Faillottaz et al., 2015

Results – disasters in remote locations (Uttarakhand, India, 7 February 2021)

At 10:30 local time a major debris flow was witnessed and shared on social media, hitting a hydropower plant under construction

By 19:00 local time satellite imagery of the event had become available that allowed us to document the reason for the event

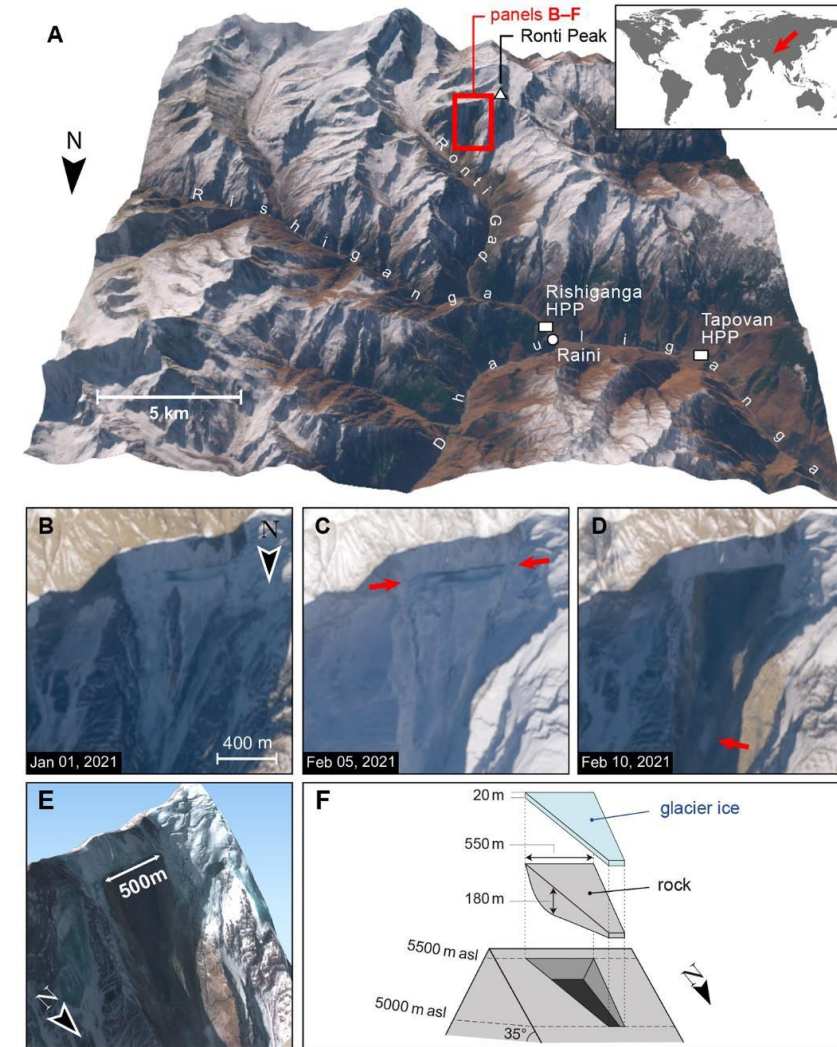
>200 people dead

Results – Utrakhand rock fall/debris flow

rockfall mixed with ice

mobilization of deposited debris

Transition into debris flow

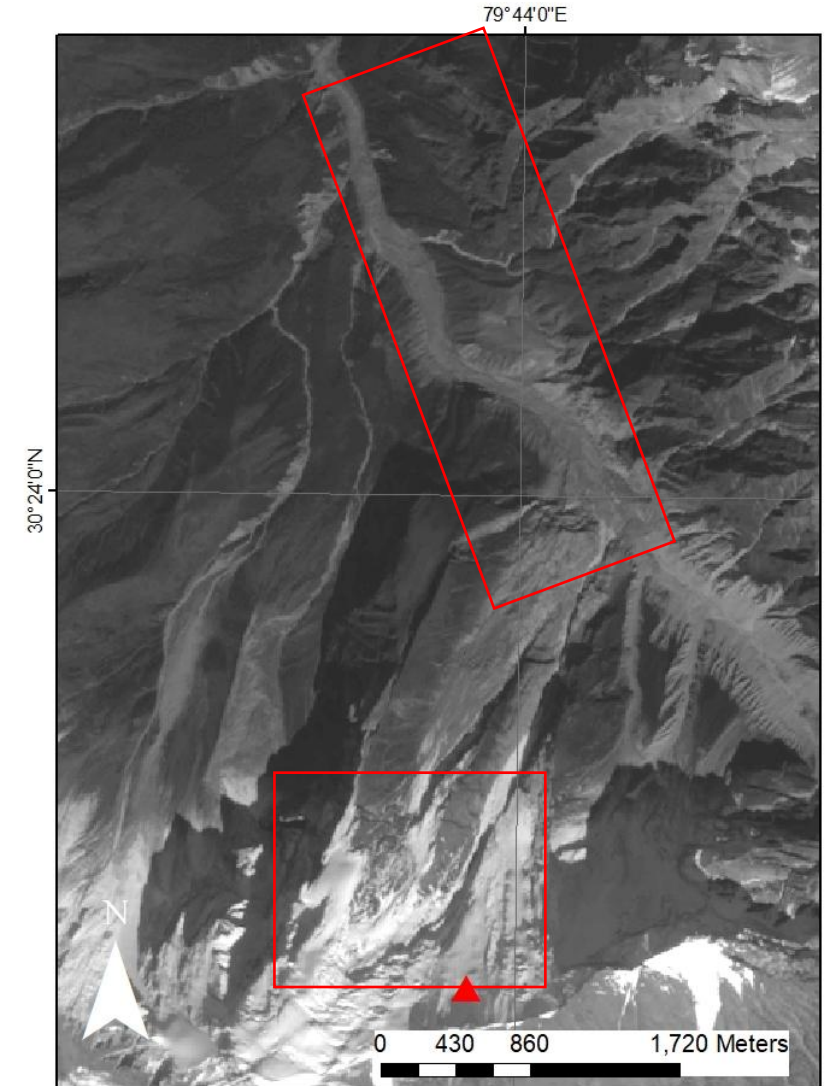


Shugar et al., 2021

Climate change? Bad luck?

Previous ice avalanche in 2016

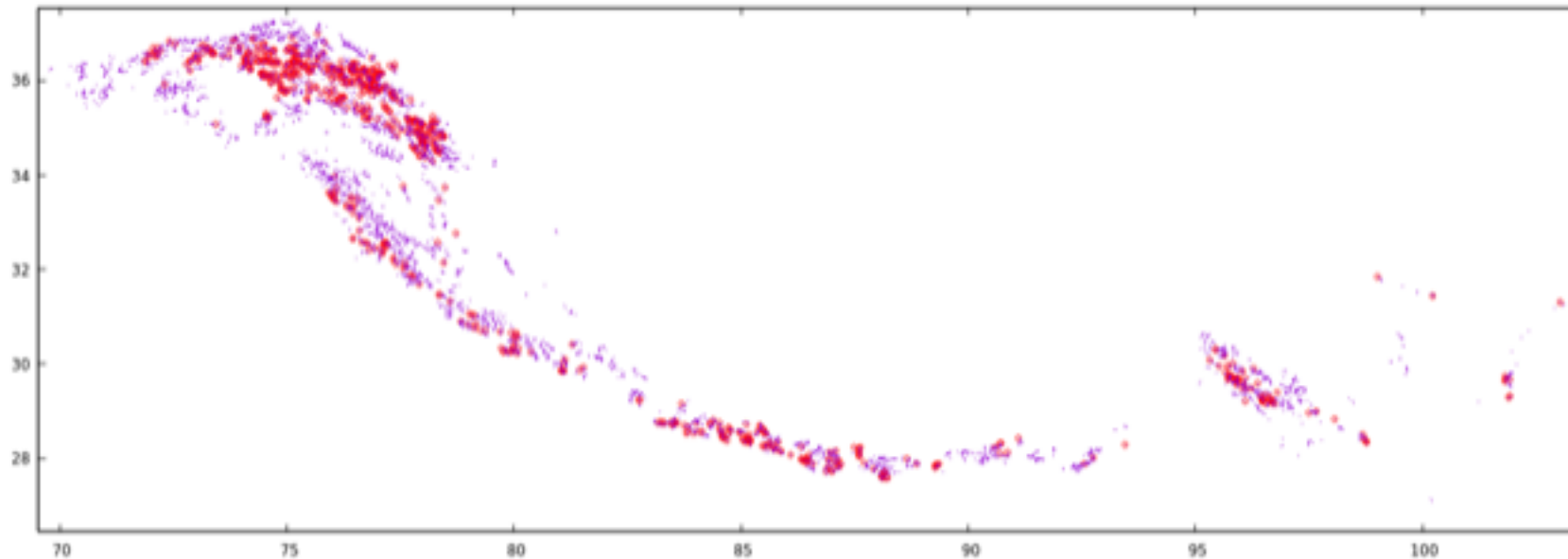
Ice/debris deposited in valley below





Is scaling this up possible?

Hanging glaciers in the Hindukush – Himalaya
- steep slope and small size



Laha et al., 2024

[Home](#) > [Natural Hazards](#) > [Article](#)

Assessing potential risk of glacier avalanches to hydropower infrastructure in the Himalayan region

Original Paper | Published: 27 January 2024

Volume 120, pages 4749–4774, (2024) [Cite this article](#)

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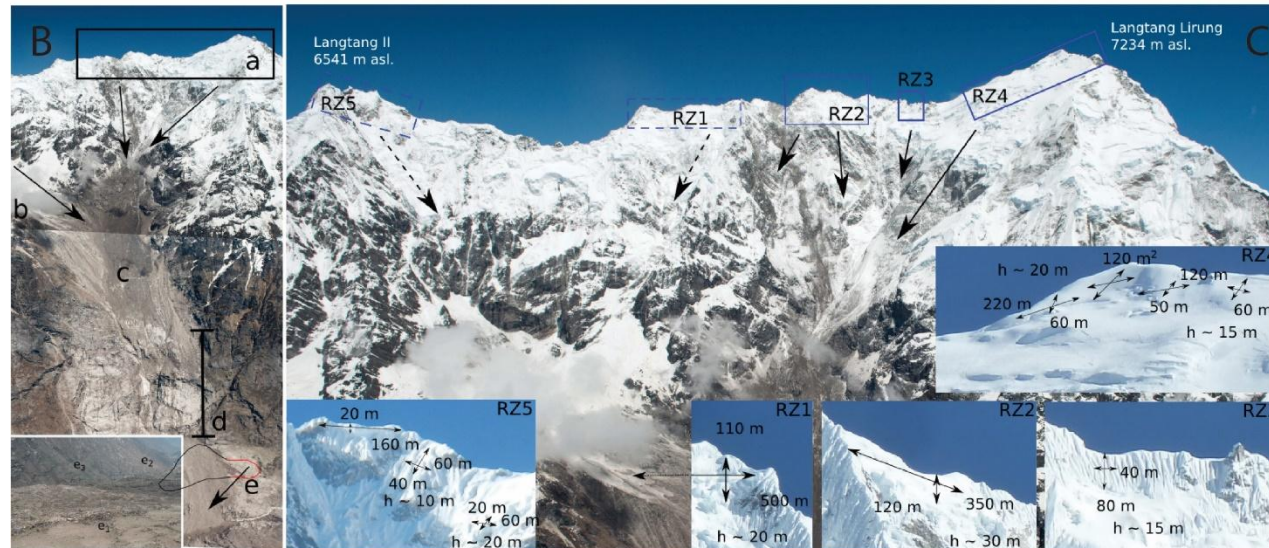
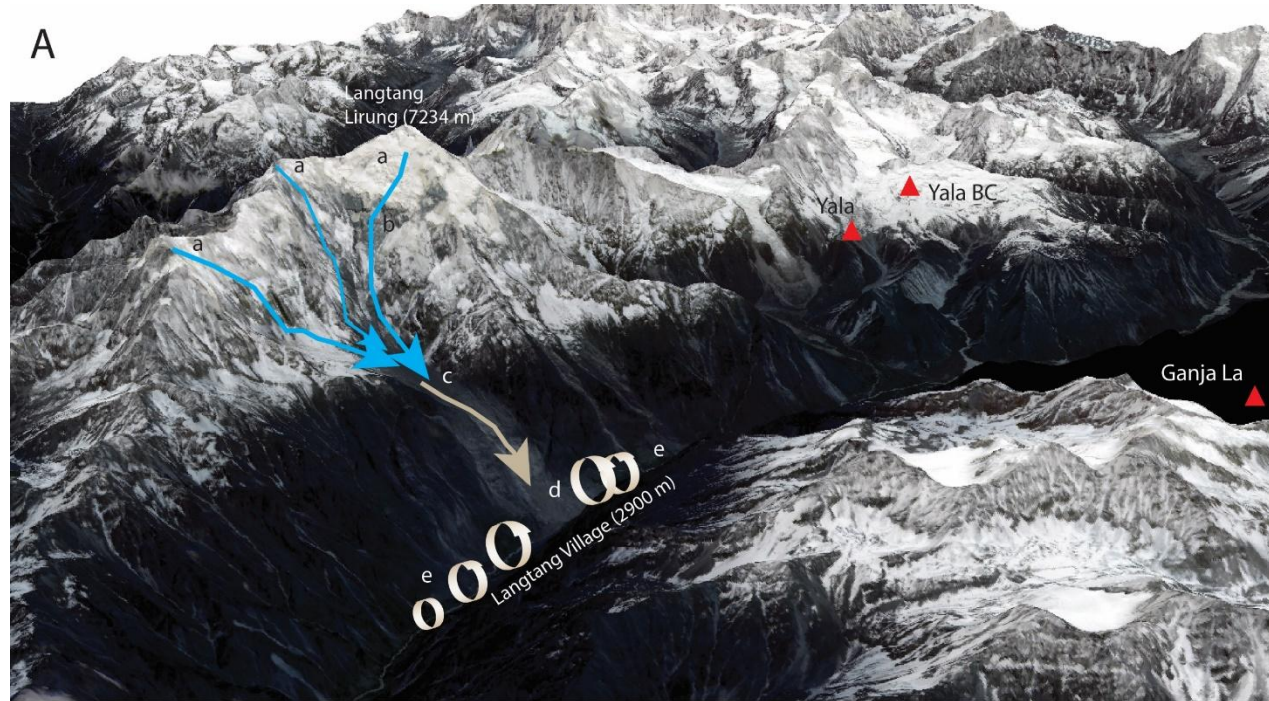
[Figures](#)

Results – co-seismic events

- Gorkha earthquake 2015 in Nepal
- >4000 co-seismic landslides
- Ice-break offs that caused deposits of up to 100 m depth
- Multi-hazard (ice break off + avalanche + debris flow)

Kargel et al., 2016





Results – co-seismic events

Zhuang et al. 2024

communications earth & environment

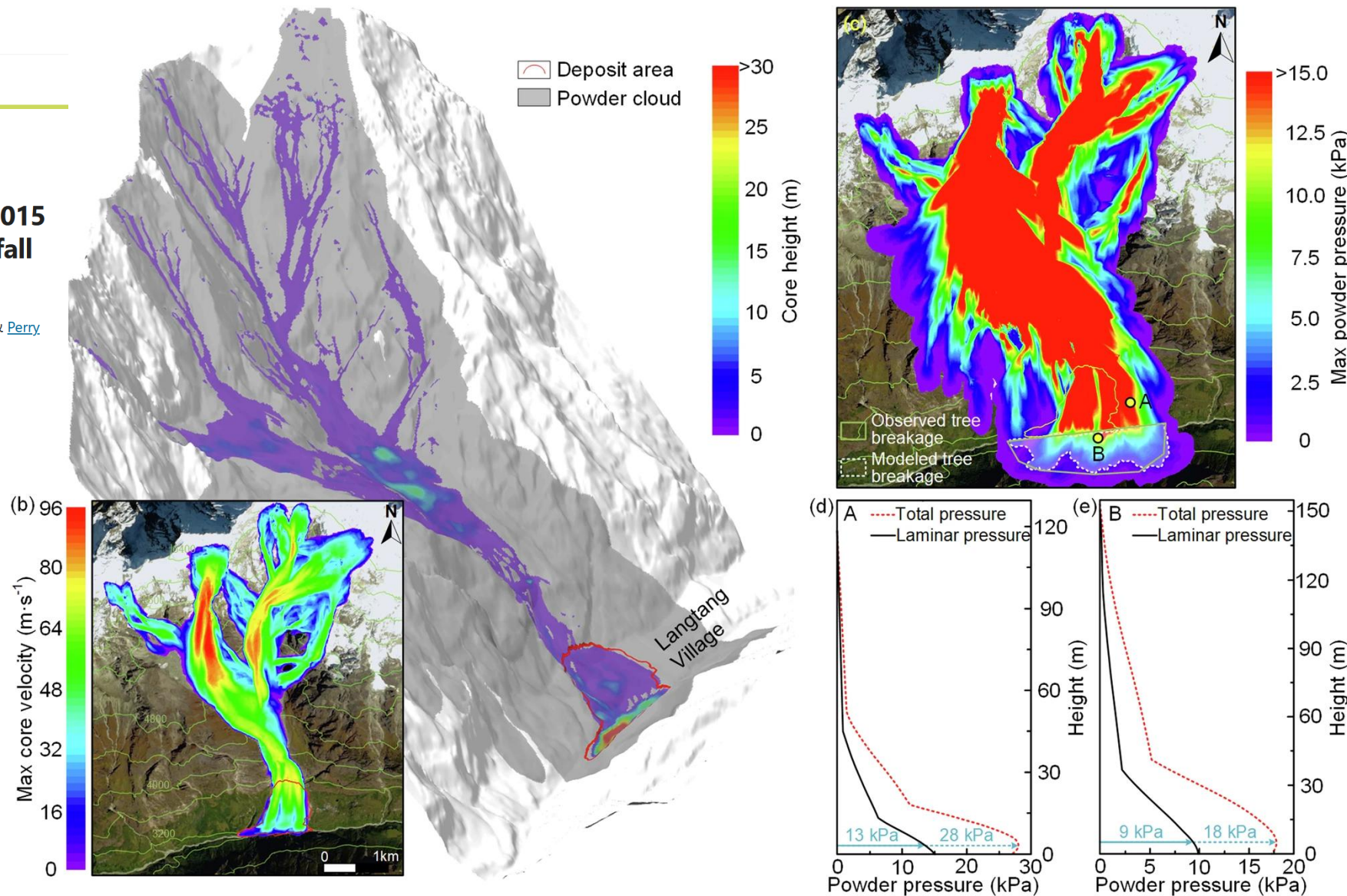
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Article | [Open access](#) | Published: 27 August 2024

An earthquake-triggered avalanche in Nepal in 2015 was exacerbated by climate variability and snowfall anomalies

[Yu Zhuang](#) , [Binod Dawadi](#), [Jakob Steiner](#), [Rajesh Kumar Dash](#), [Yves Bühler](#), [Jessica Munch](#) & [Perry Bartelt](#)



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... what appears when the ice is gone?

Kaab et al., 2005

Glacier length variations surges

Glacier tongues extend within days to weeks by 100s of meters

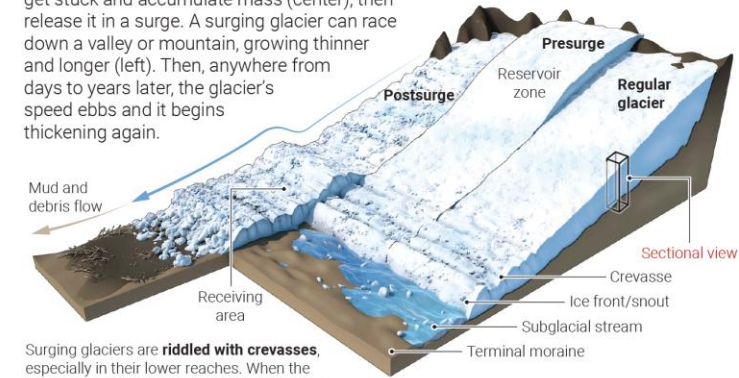
Decadal return period

No net mass gain or loss

Unclear why these events happen – no apparent link to changing climate

A glacier unleashed

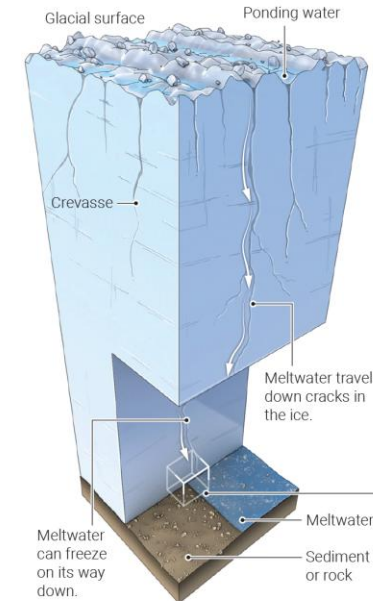
Glaciers gain mass in their upper reaches, where snowfall is heavier, and lose it at their snouts, where the ice breaks up and melts (right). Most glaciers flow steadily, but some get stuck and accumulate mass (center), then release it in a surge. A surging glacier can race down a valley or mountain, growing thinner and longer (left). Then, anywhere from days to years later, the glacier's speed ebbs and it begins thickening again.



Surging glaciers are riddled with crevasses, especially in their lower reaches. When the surge ends, meltwater that built up under the glacier before the surge may sweep mud and debris from its snout.

Trickling down

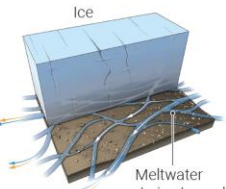
Meltwater plays a key role in triggering surges. Pooling on the glacier's surface, it can seep down into crevasses. There it can refreeze, releasing heat that softens the ice; it can also pool at the base of the ice.



Steady state

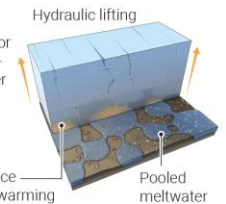
In a "normal" glacier, meltwater drains efficiently from its base, carrying away heat and leaving the ice anchored to its bed.

Steady meltwater flow under glacier releases heat and pressure.



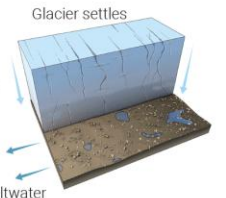
Buildup to a surge

If drainage is poor or melting accelerates, meltwater can accumulate under a glacier, warming the ice and lifting it off the ground.



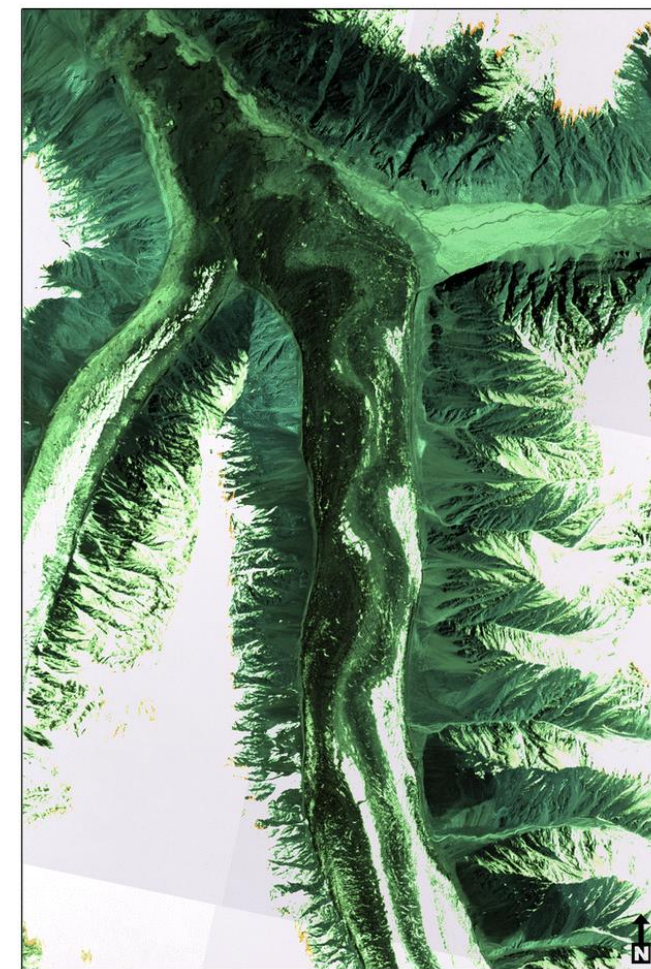
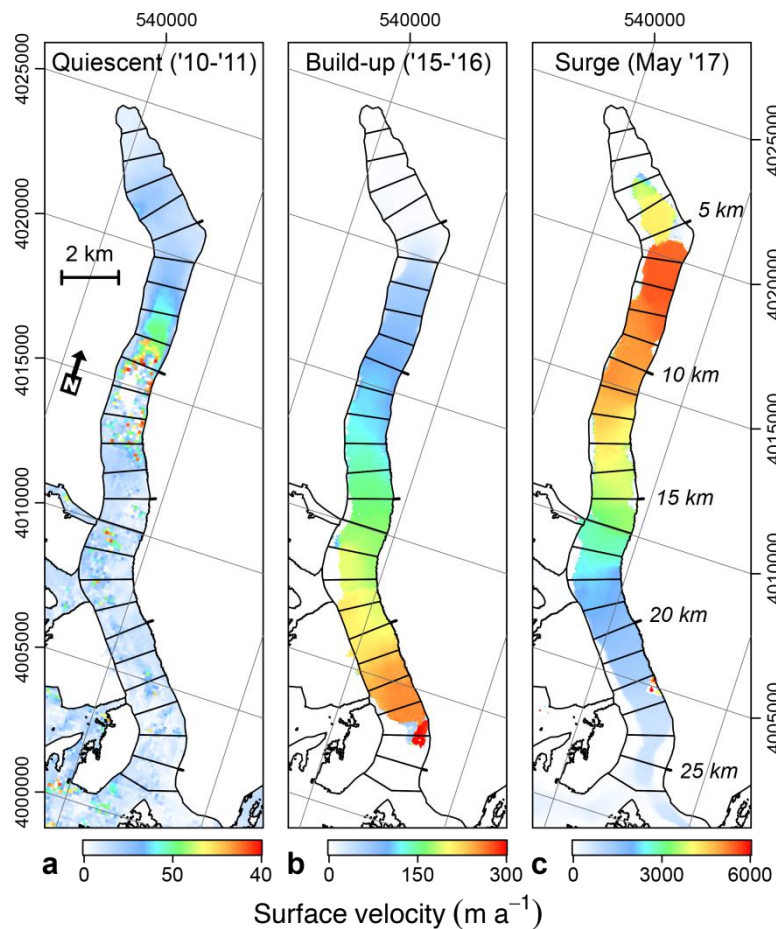
Aftermath

Once the surge releases the meltwater, the glacier subsides onto its bed, and the cycle begins again.



Surge in Karakoram 2017

Glacier velocities from remote sensing



Steiner et al., 2018, Cryosphere

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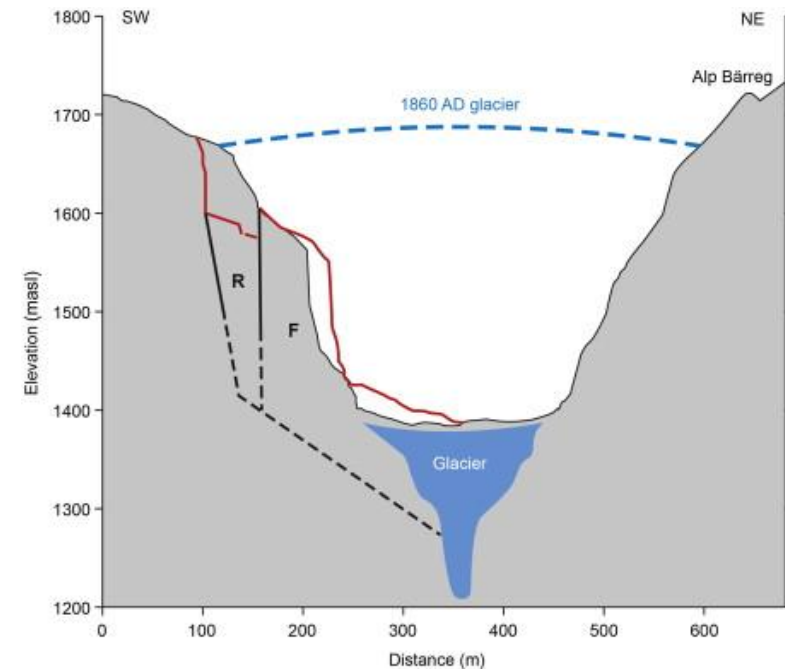
Destabilization of frozen soil

Increased rock slope failure in post-glaciated environments

Dependent on geology

Freeze-thaw within rock cracks

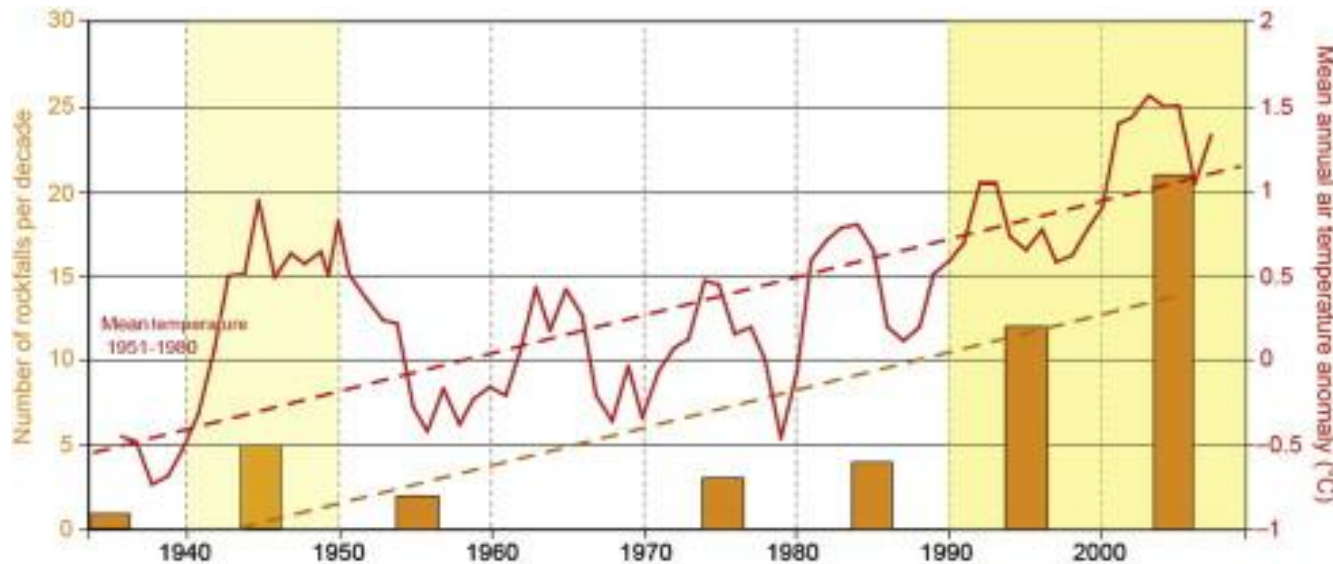
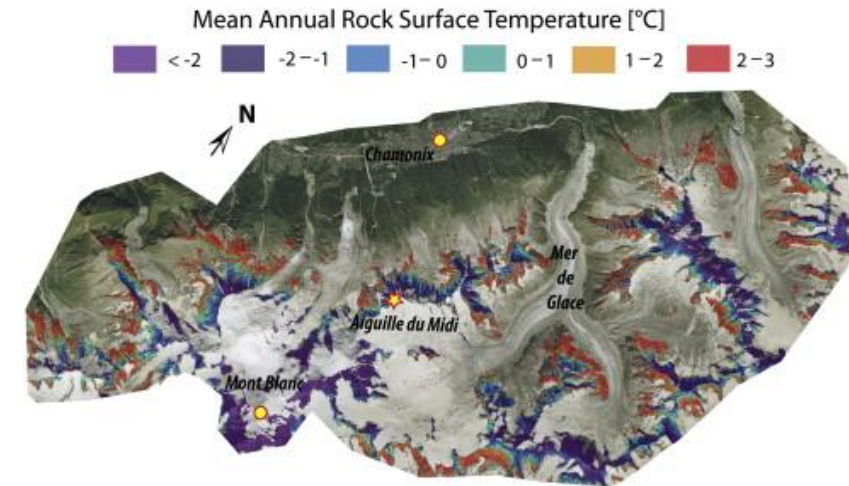
Isostatic rebound after deglaciation



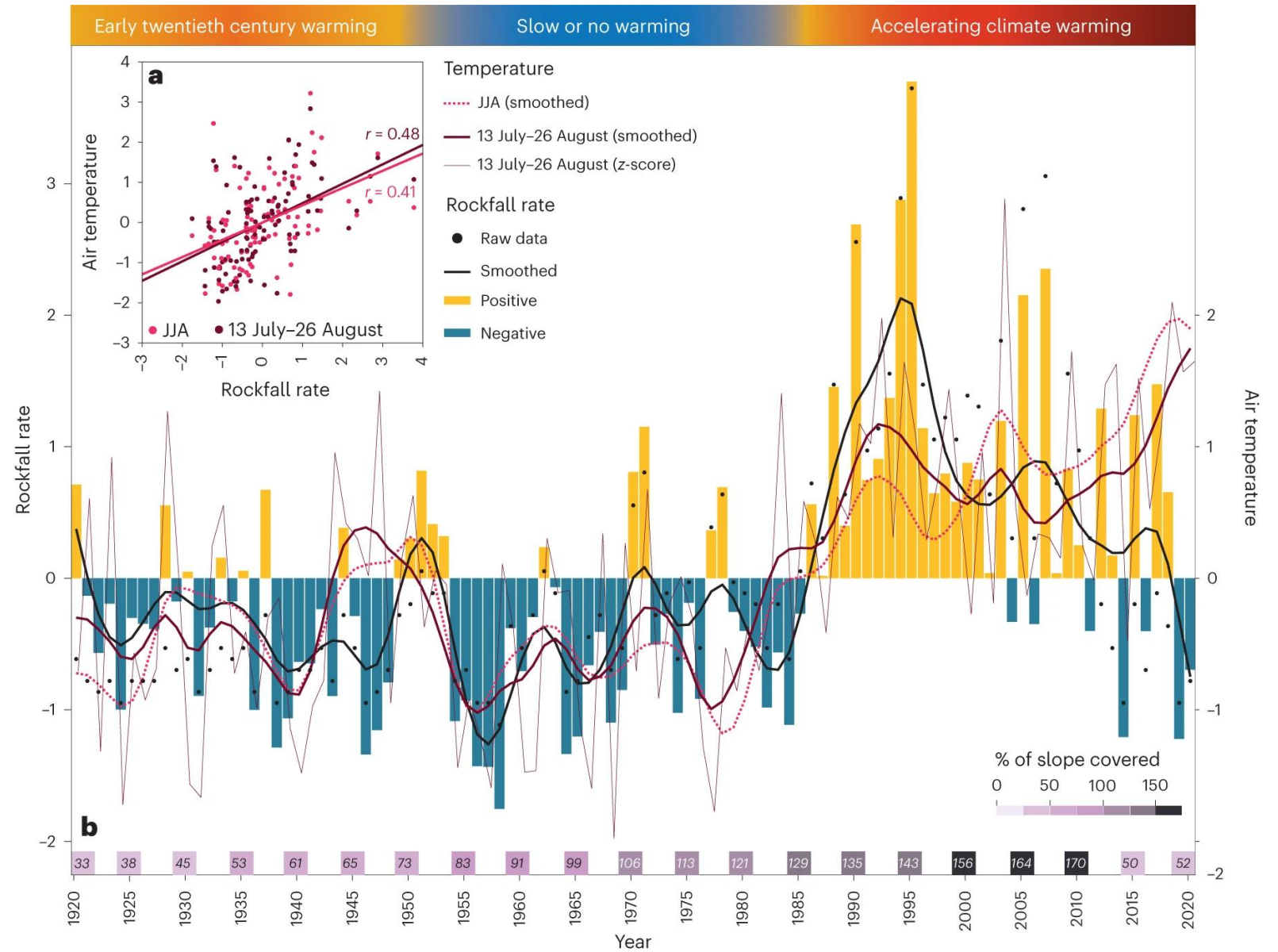
Deline et al., 2015

Destabilization of frozen soil

- Likely direct link with increasing temperatures



Deline et al., 2015



Glacier related hazards

glacial lake outburst floods (+ resulting events)

ice break offs (+ resulting events) / collapsing glaciers

glacier length variations

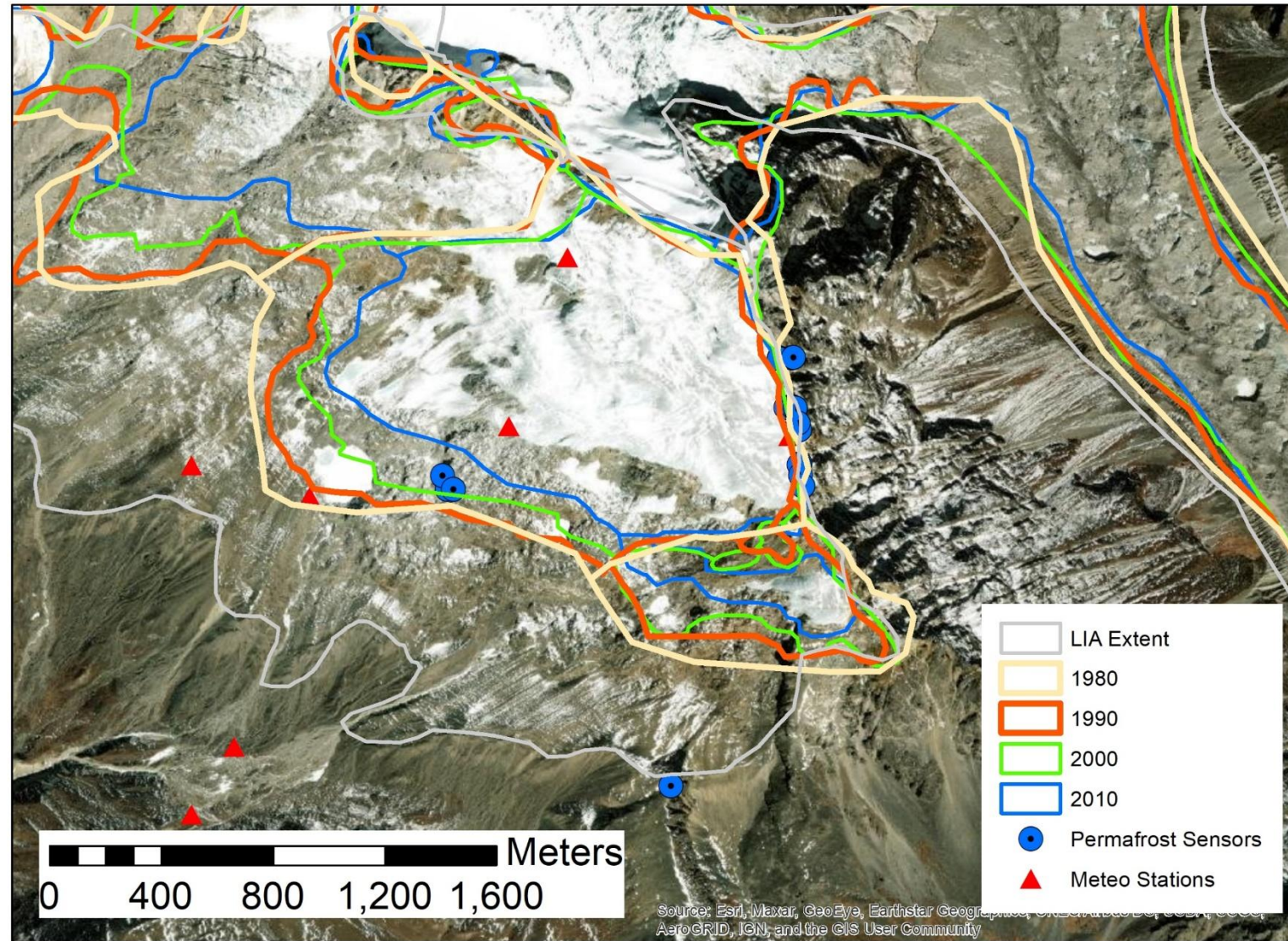
destabilization of frozen soils/slopes

... what appears when the ice is gone?

Kaab et al., 2005

Incre

- Wh



Inc

- W



Terminology

Global ice

Glacial hazards

Risk management

Case studies

Effects downstream – debris flows



Destabilization of frozen soil



Beyond the glaciers – cryosphere hazards

Snow avalanches

Snow drought

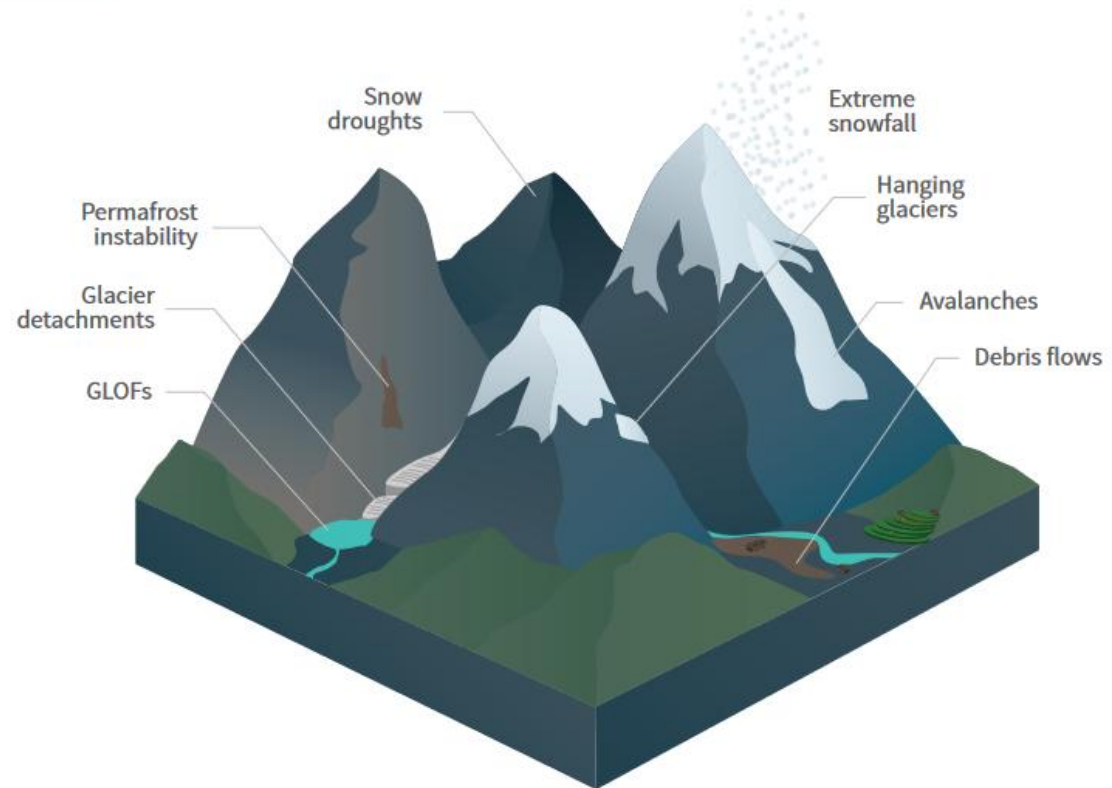
Unstable slopes due to permafrost thaw

Debris flows/landslides as a result of unstable slopes

Methane emissions as a result of permafrost thaw

FIGURE 3.5

A SCHEMATIC REPRESENTATION OF THE DOCUMENTED HAZARDS RELATED TO THE CRYOSPHERE AND HYDROSPHERE IN THE HKH



Nepal, Steiner et al.

Topics

- 1) Terminology: Hazards. Risks. Mitigation.
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- 5) *Three focus excursions:*
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 - b) The politics of national risk management in high mountains (Nepal)*
 - c) The global politics of 'glacial risks' (Greenland)*

Monitoring and hazard mitigation

Remote Sensing

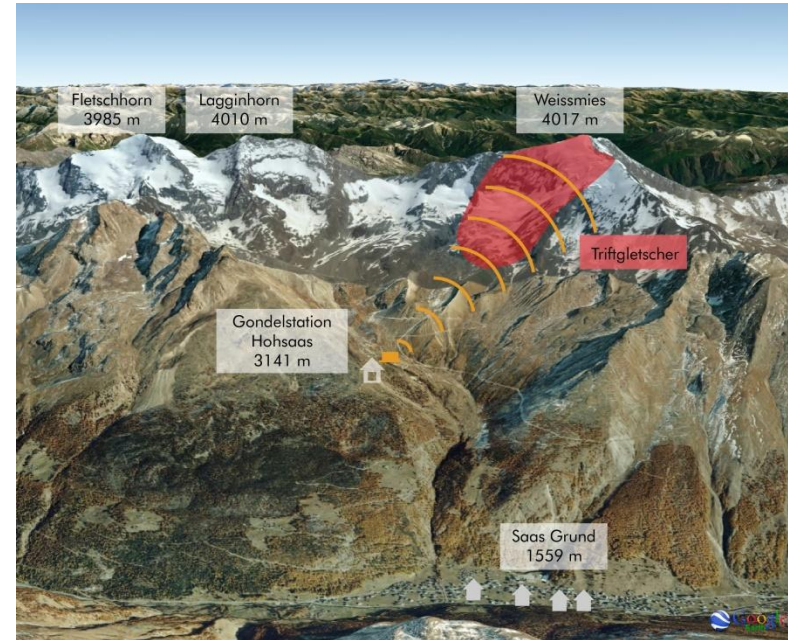
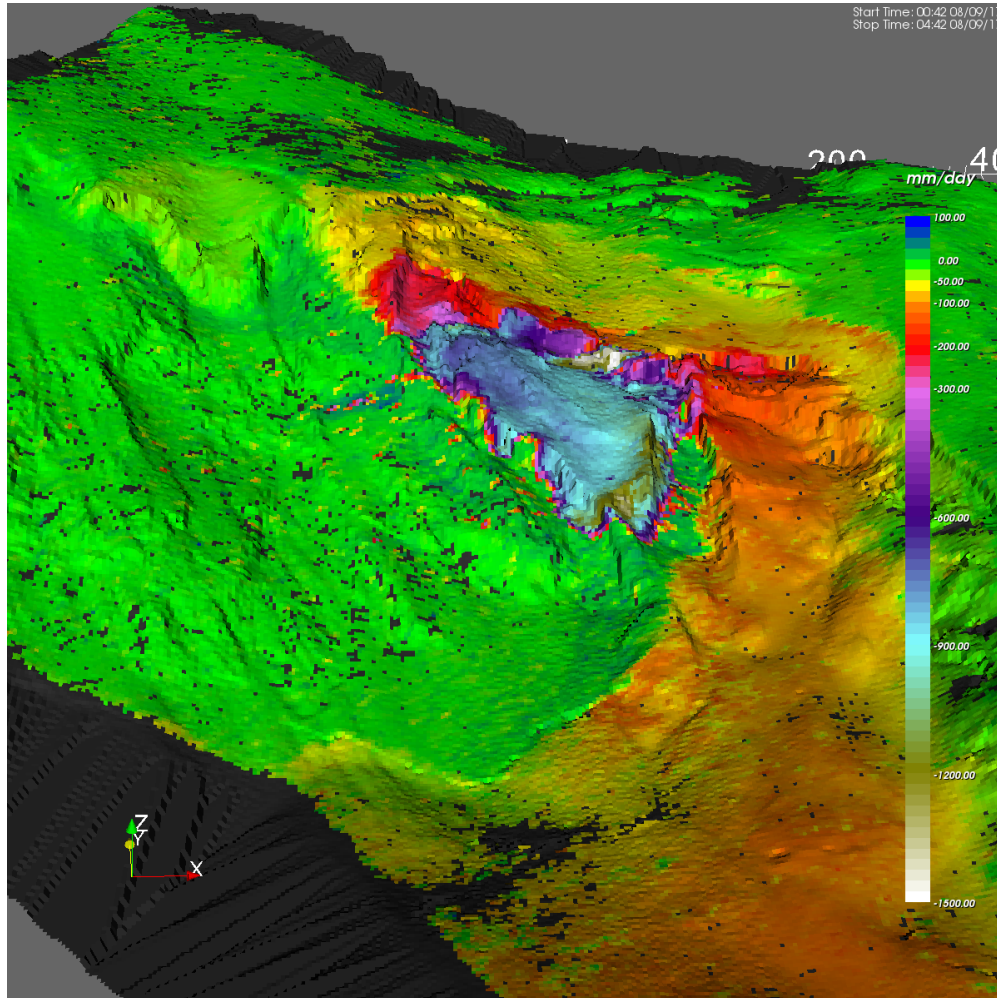
- new near real-time products for observations (Planet, Sentinel, new Radar satellites ...)
- Temperature/Precipitation products

Field based

- climate and discharge monitoring
- radar monitoring

Local knowledge?

Monitoring ice falls



Blatten, Switzerland, May 2025



Terminology

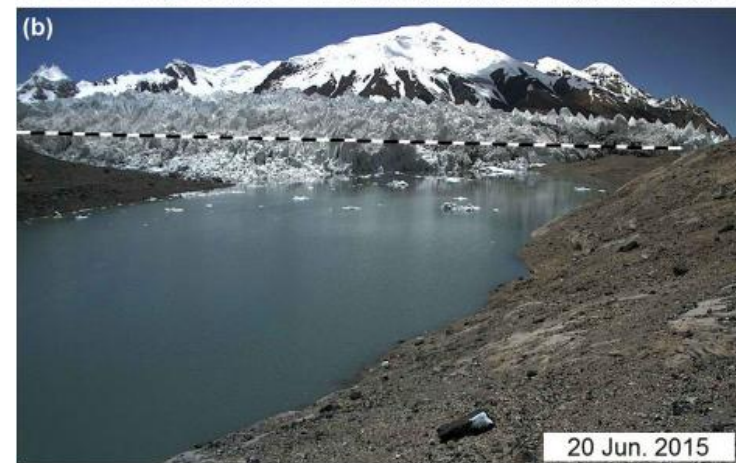
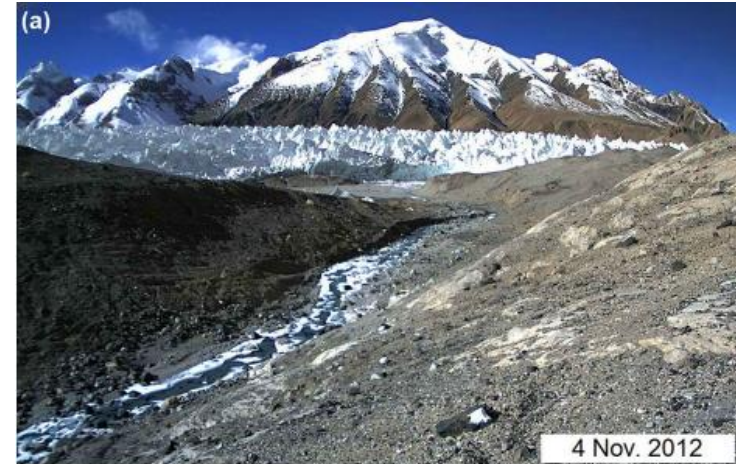
Global ice

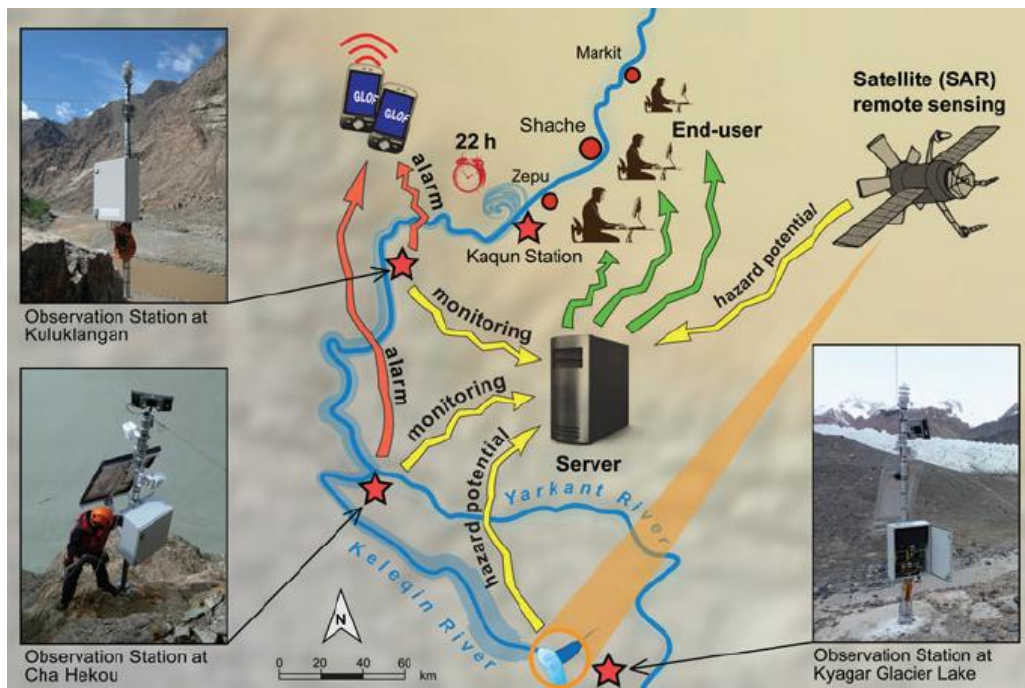
Glacial hazards

Risk management

Case studies

Monitoring a glacier lake





Mapping and modelling

- The basics not any different from elsewhere
- Extra challenge:
 - Lack of data
 - Difficult topography for satellite imagery as well as numerical modelling
 - Access to affected populations difficult (including language/cultural barriers)

Hazard Mapping

Remote sensing

- Maps of potential instable areas
- Identification of weak moraines/ice cores/rapid surface changes on glaciers
- Slope maps + exposure to solar radiation (freeze thaw cycle)
- Exposure maps

Field based

- Runout lengths
- Damage classification (pressure, inundation)
- Interviews

Impact - observed



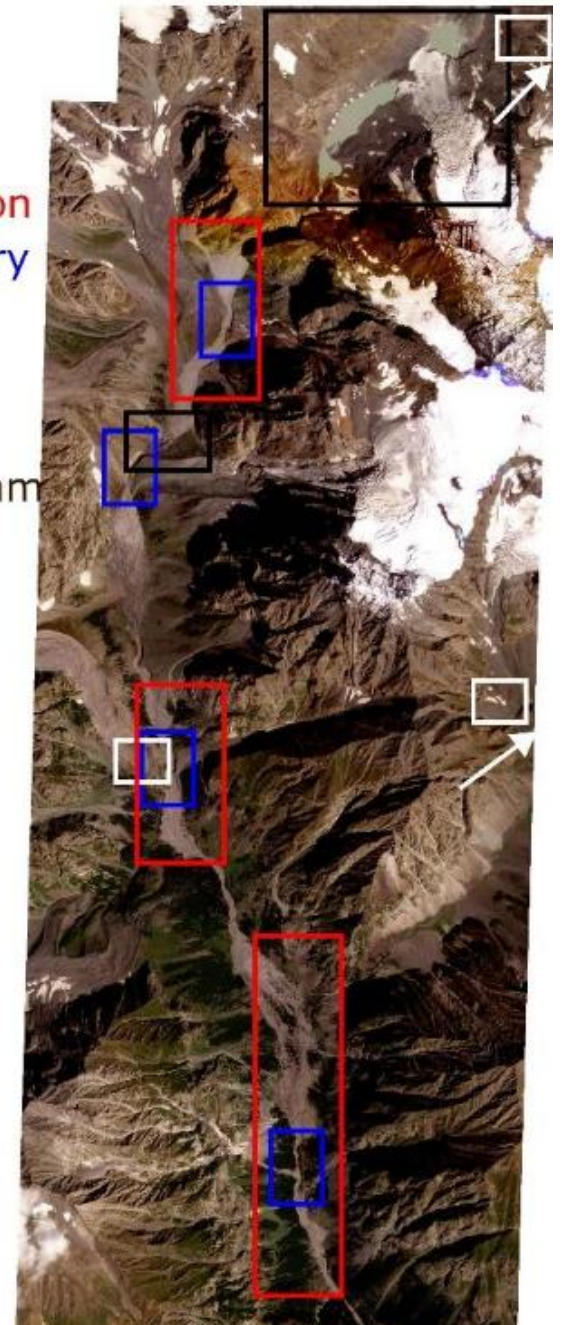
drained lake

significant deposition
significant secondary
erosion

Shazore lake and
likely temporary dam

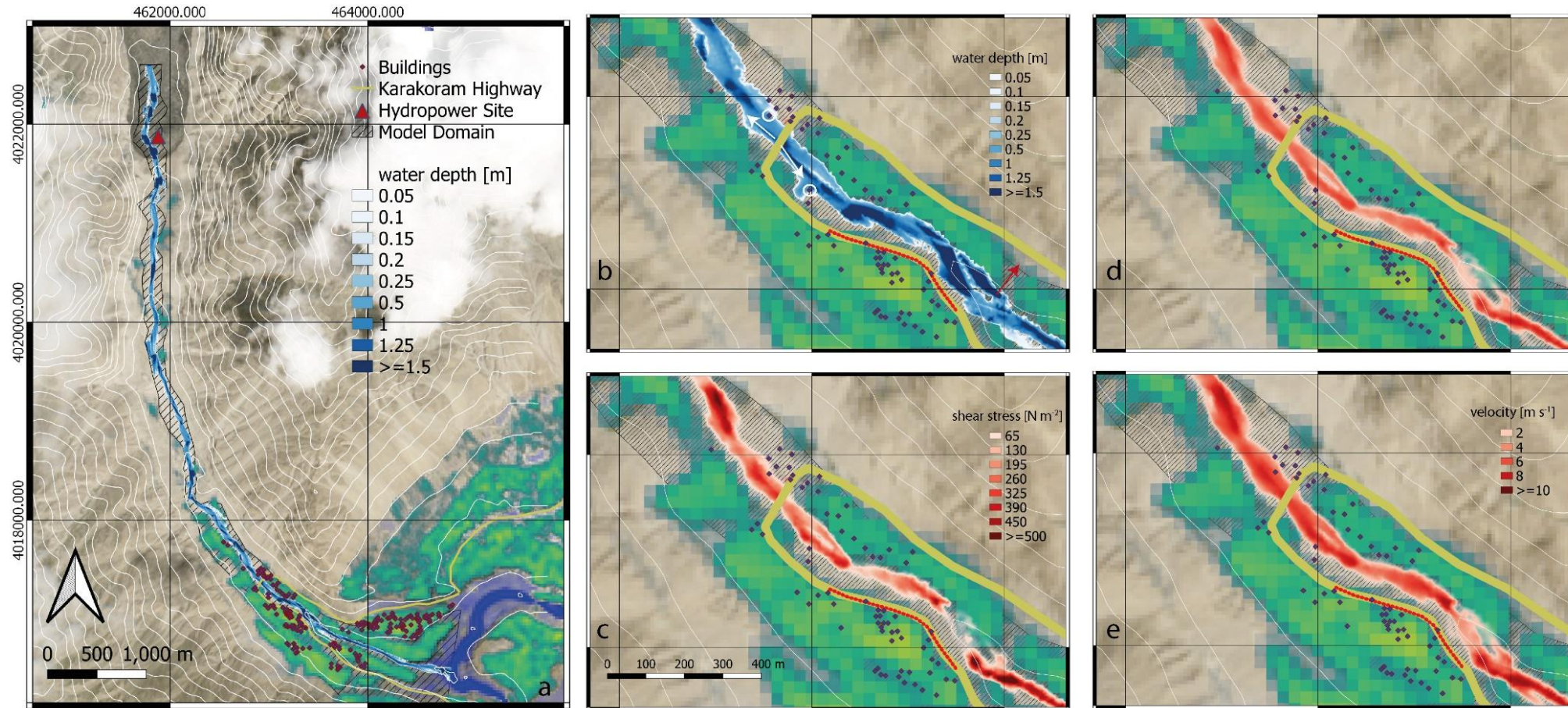
additional glacial
lakes

0 750 1,500 m



<https://hirisk.org/risk-assessments/>

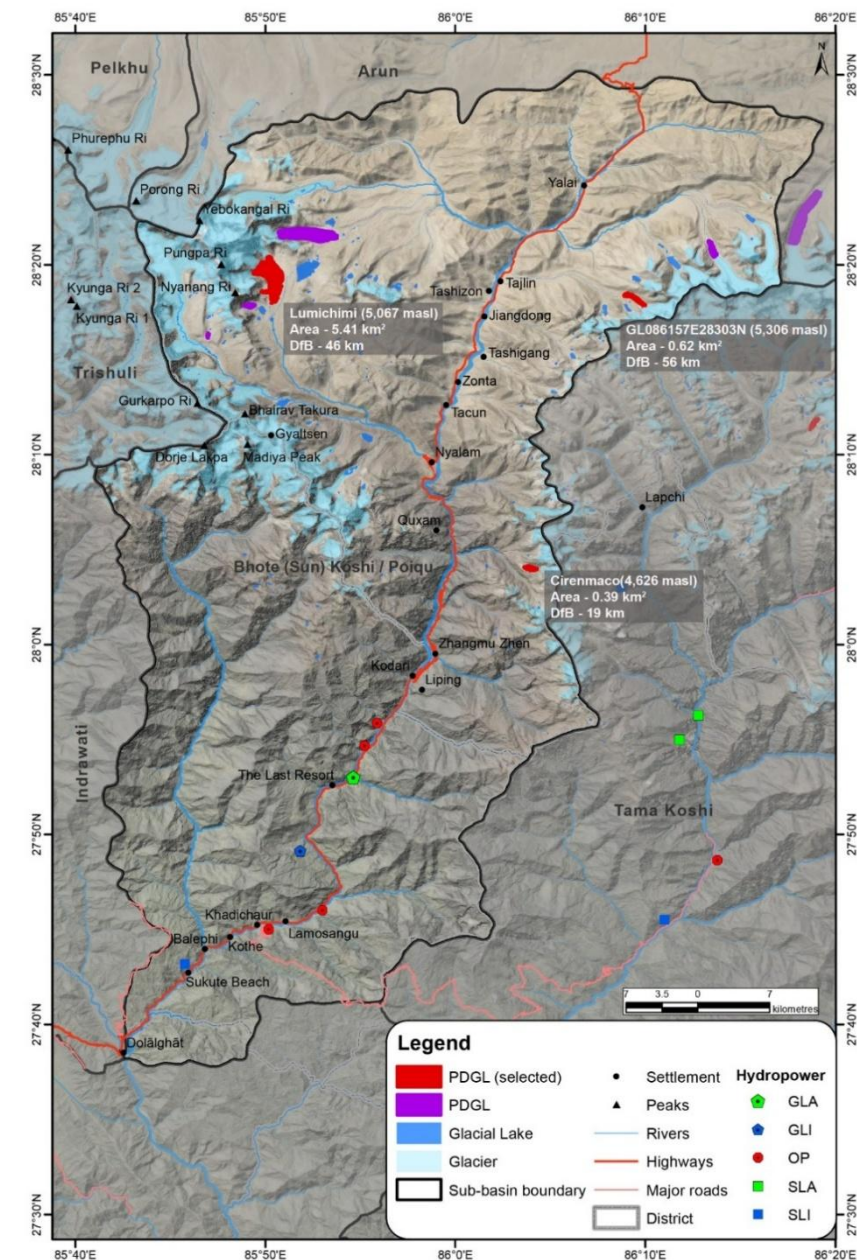
Impact - modelled



Muhammad et al., 2021

Challenge across borders

- Transboundary multi-hazard and risk assessment
- China-Nepal border



Local and Indigenous knowledge

- Focus Group Workshops
- Interviews
- Awareness Raising



Shimshal, Pakistan



Melamchi, Nepal

Local and Indigenous knowledge

- How do we turn local knowledge into (western) scientific data?
- To whom do glaciers matter, how and when?

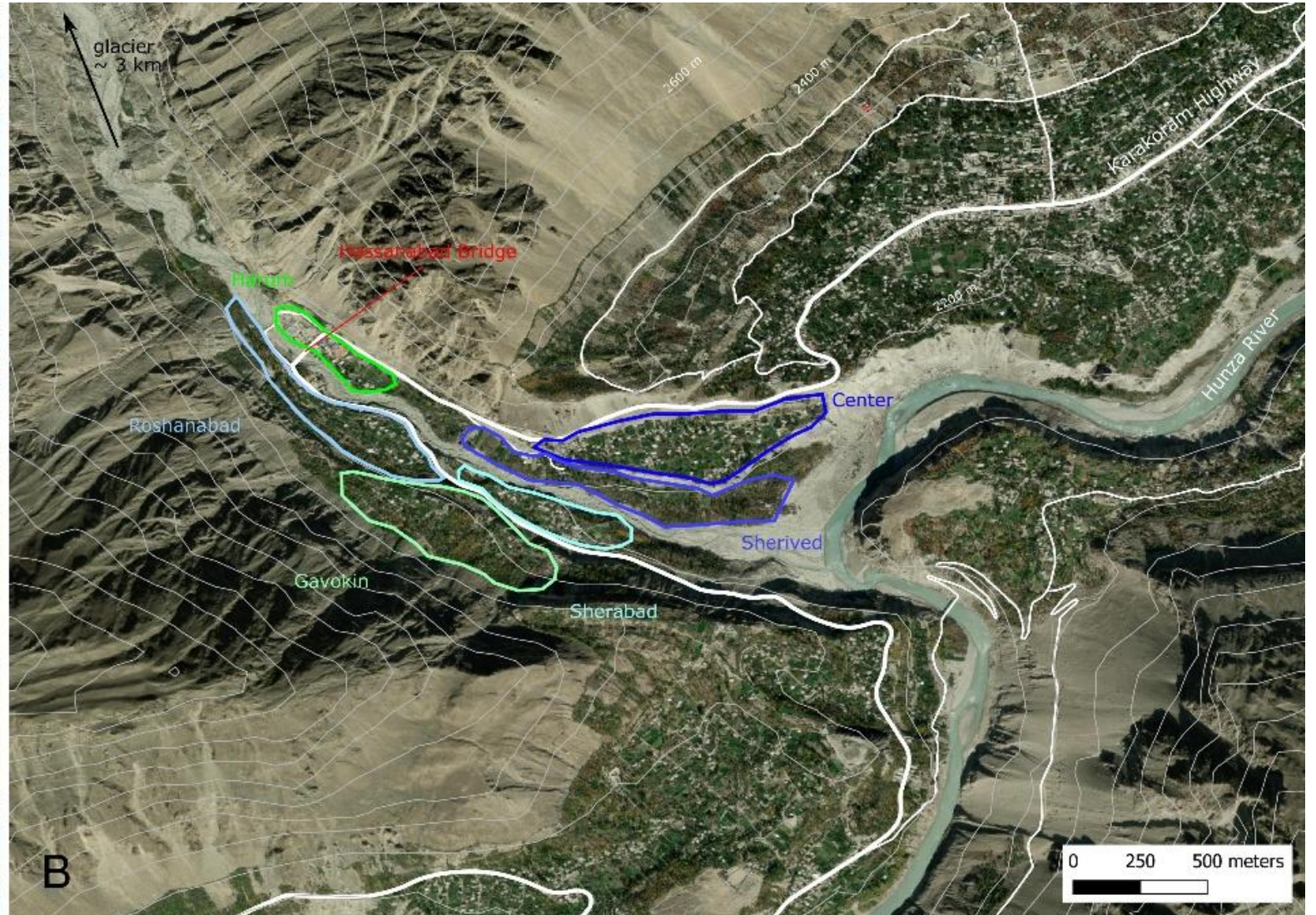
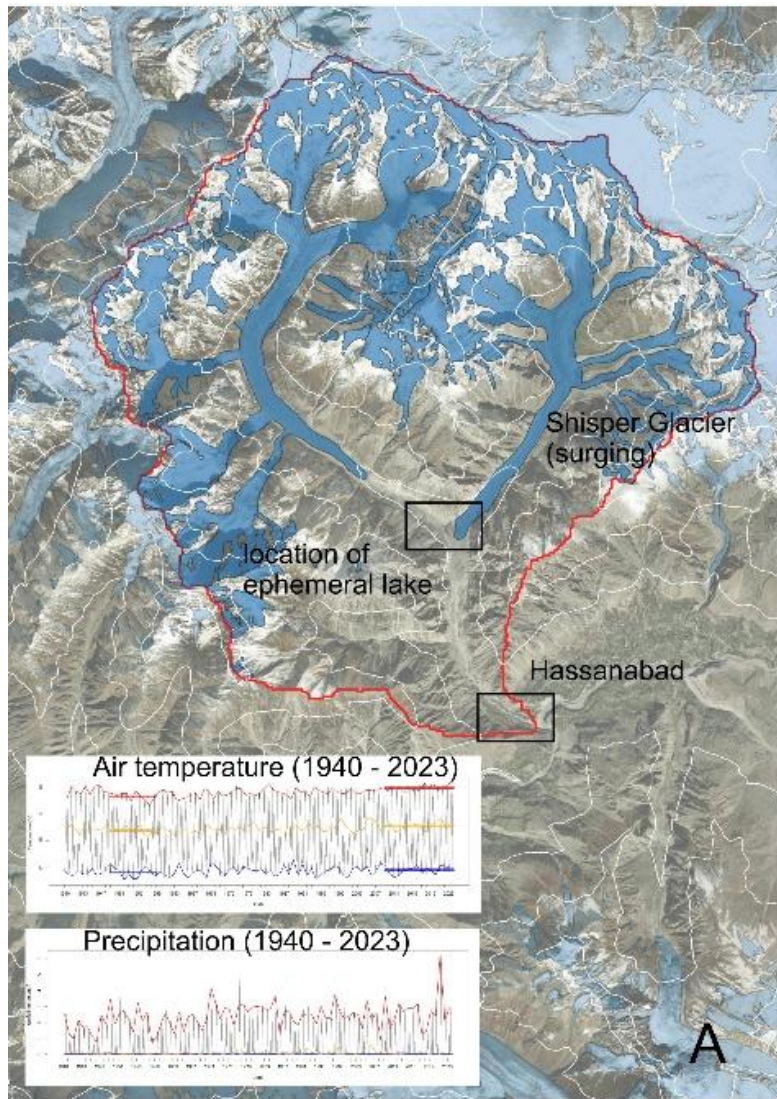


Acharya et al., 2023

Topics

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 - c) *The global politics of 'glacial risks' (Greenland)*





Aaliya et al., (in preparation)

Table 9. Prevalence of mental health outcomes relative to the number of GLOF events experienced

Number of GLOF Events experienced	Total (n)	Anxiety_Cases (GAD-7 $\geq$ 10)	Depression_Cases (PHQ-9 $\geq$ 10)	PTSD_Cases (IES-R $\geq$ 33)	Anxiety PrR (%)	Depression PrR (%)	PTSD PrR (%)
1	14	1	0	5	7.14	0	35.71
2	18	2	3	11	11.11	16.67	61.11
3	20	3	1	12	15	5	60
4	213	34	38	122	15.96	17.84	57.28

Aaliya et al., (in preparation)

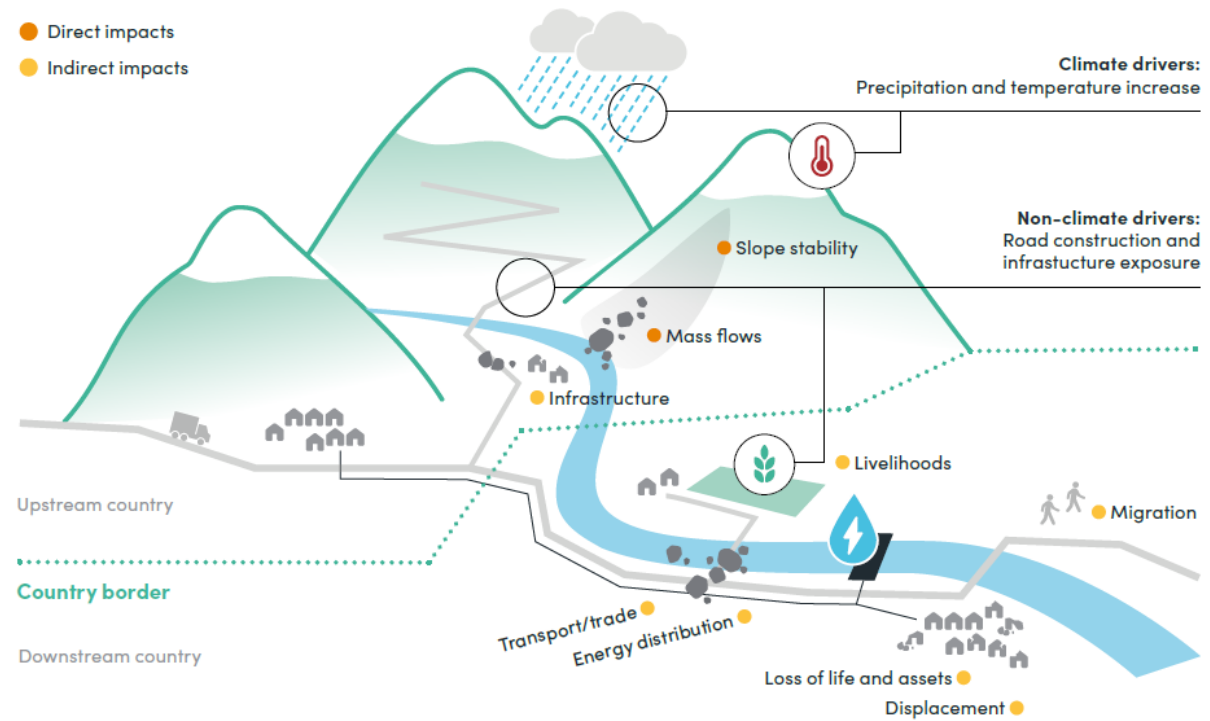
- How do we disentangle impacts from individual events from slow onset changes?
- How do we disentangle climate related stressors and those coming from institutional/governance side?



Topics

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Risk



Steiner et al., 2023

Upstream – downstream linkage

Cascading hazards and risks (*knock on effects*)

Transboundary risk

纬度：28.402561°N — 经度：85.646705°E — 海拔：5160米

日期：2025年7月7日至8日

当地时间：未知（首次报告的影响时间为0530中国标准时间/0315尼泊尔标准时间）

灾害类型：冰川湖溃决洪水（GLOF）

影像资料：（来源：Sentinel-2，2024年6月至2025年7月）

GLOF in August 2025

20 people killed

Border post between China and
Nepal destroyed and subsequently
closed for half a year

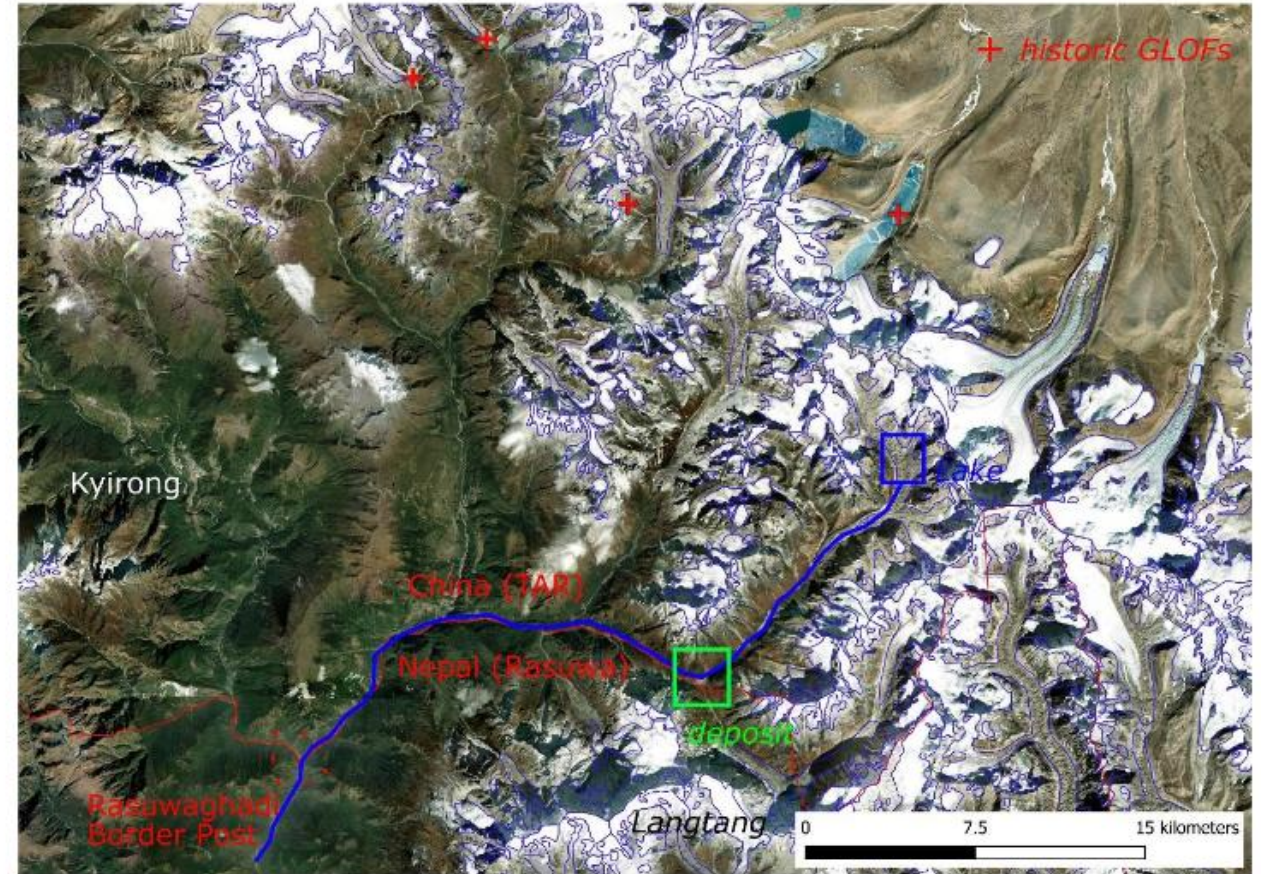


图1：从Purepu冰川到中尼吉隆口岸-尼泊尔Rasuwaghad的完整GLOF路径

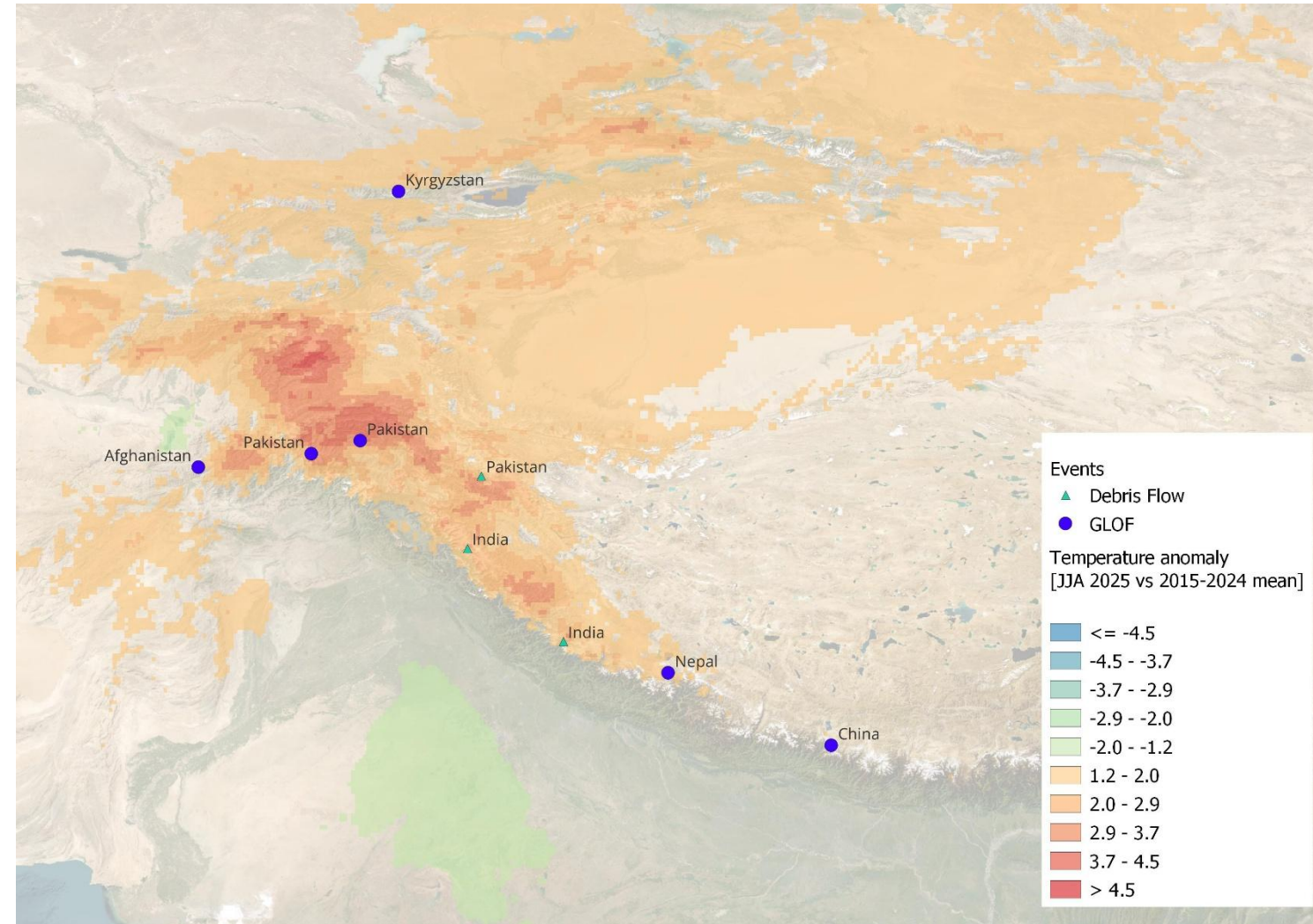
Khatri, Steiner et al. (working paper)

GLOF in August 2025

20 people killed

Border post between China and
Nepal destroyed and subsequently
closed for half a year

Extreme summer heat waves at
altitude



Allen et al. (in preparation)



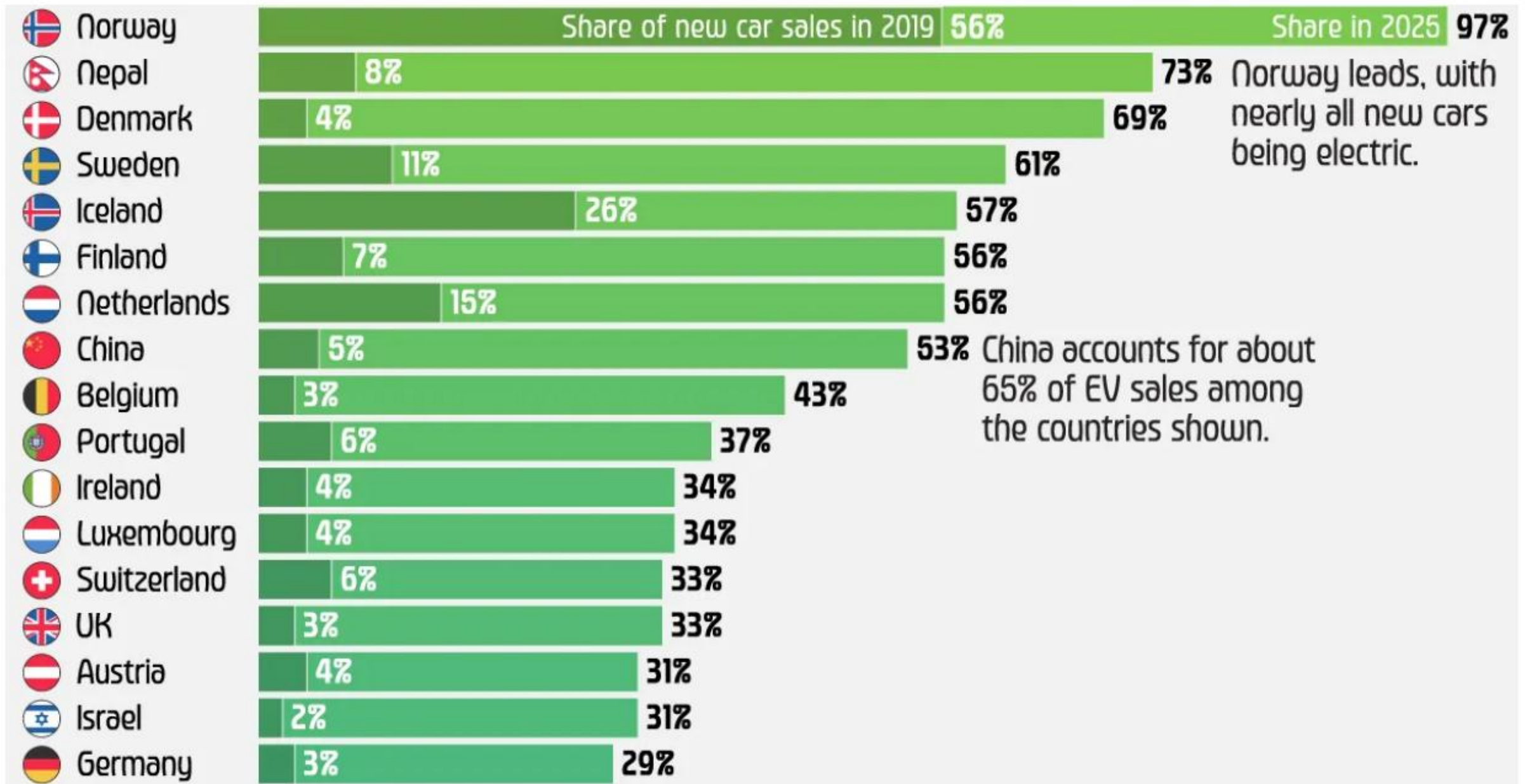
Terminology

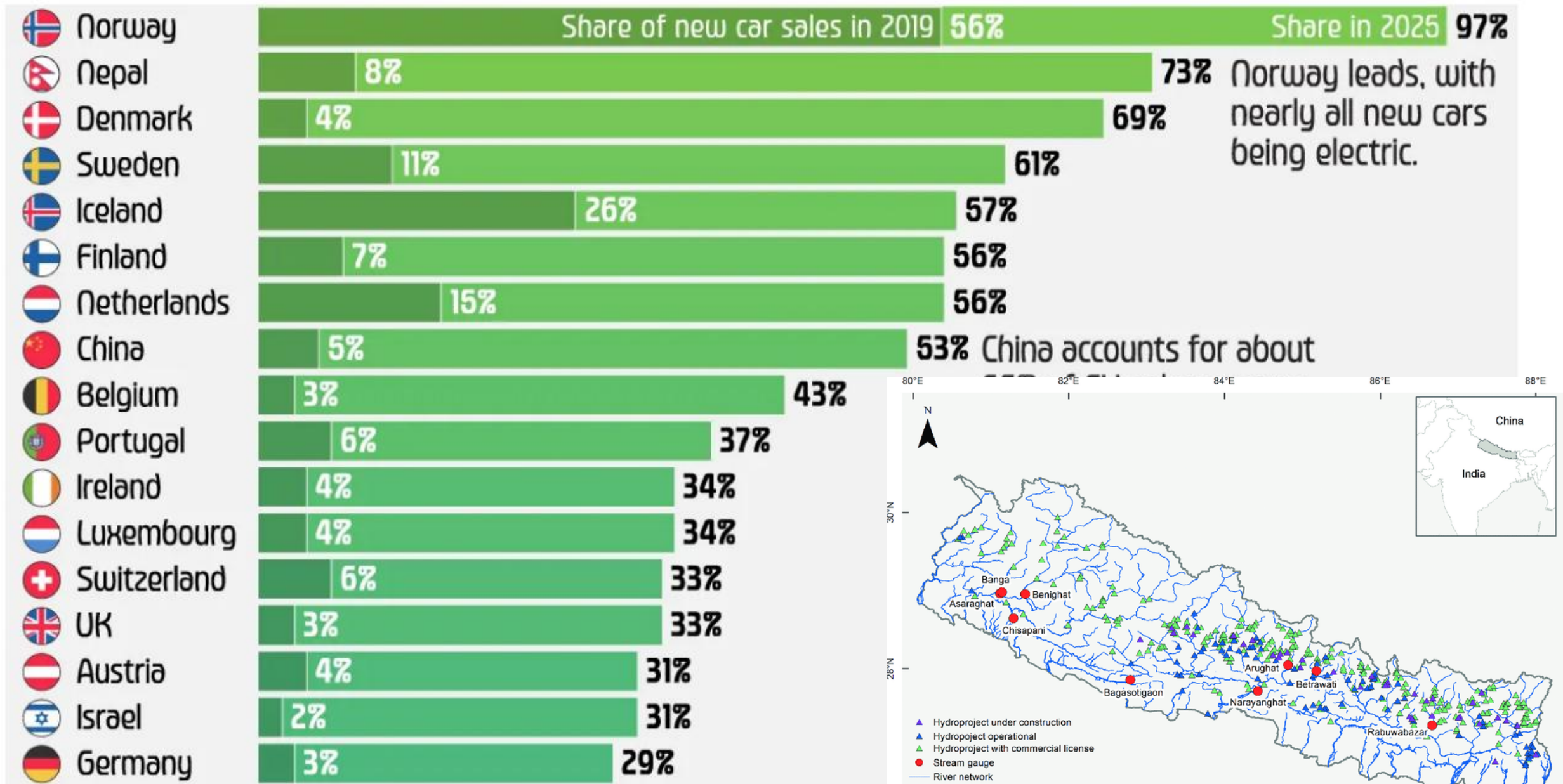
Global ice

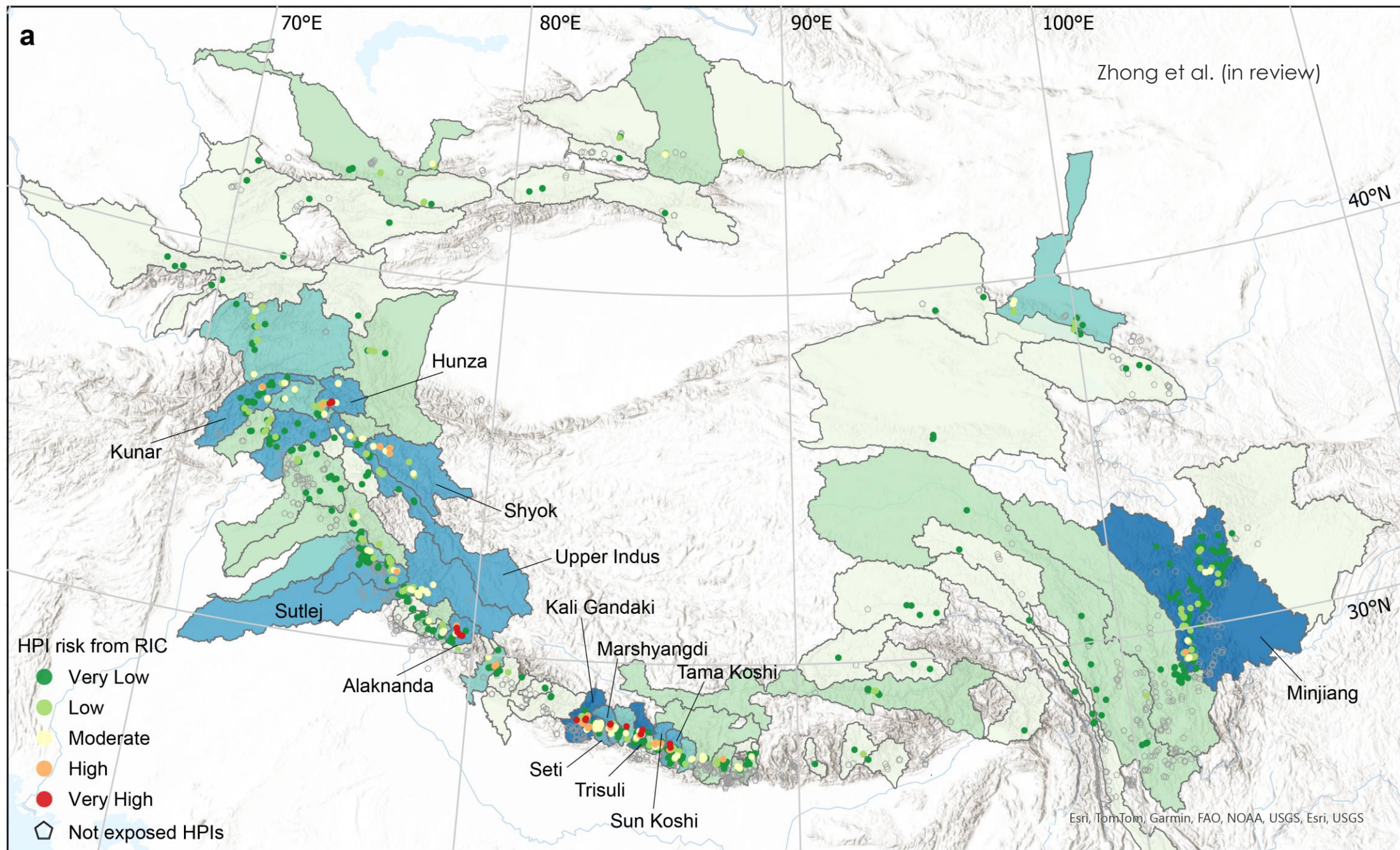
Glacial hazards

Risk management

Case studies







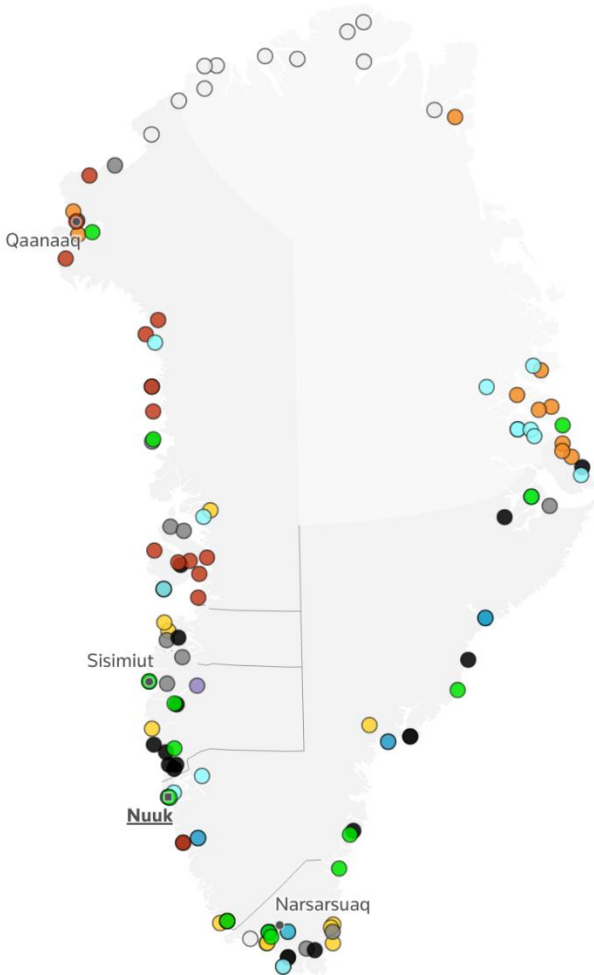
Topics

- 1) Terminology: Hazards. Risks. Mitigation.
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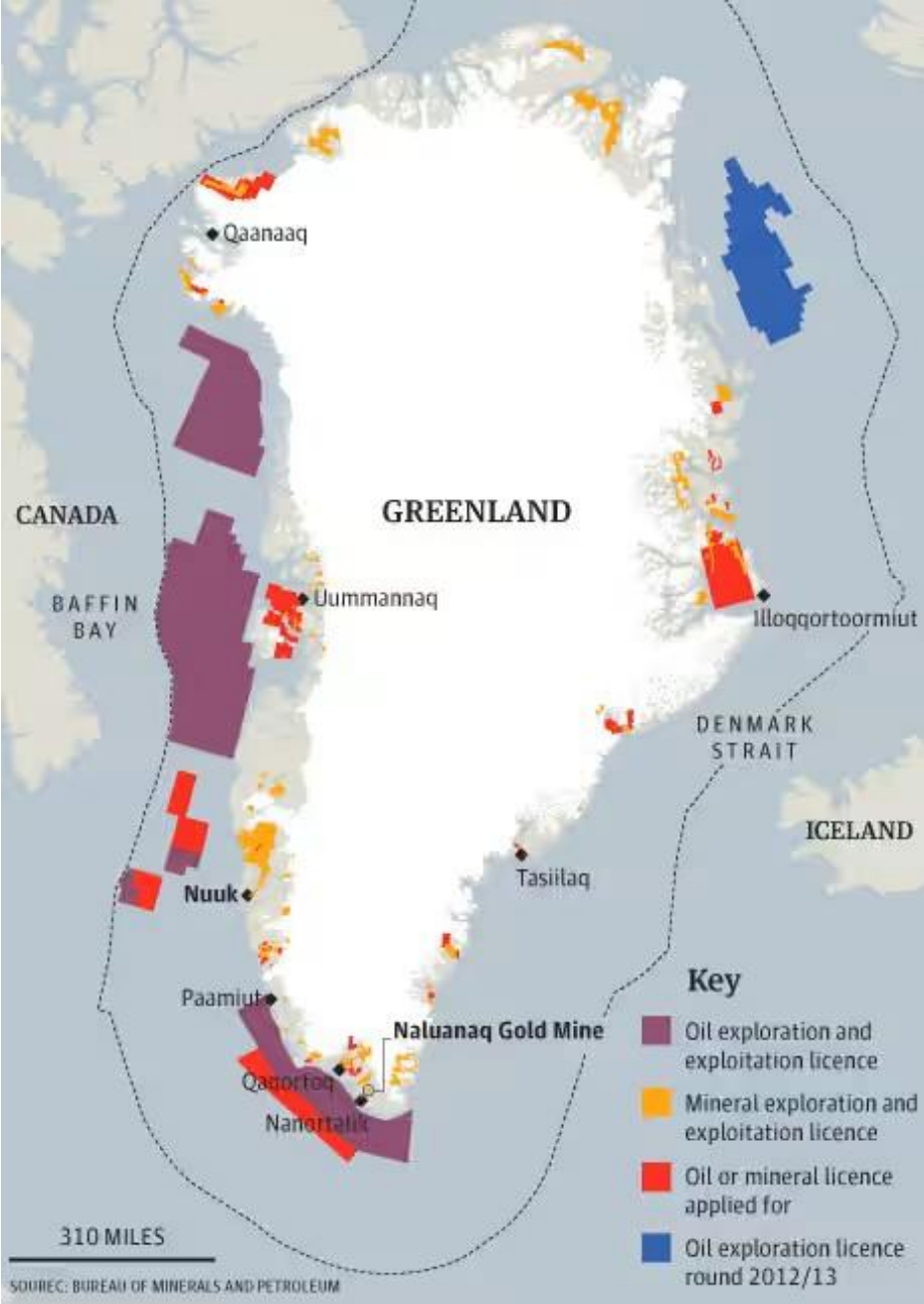
Greenland's mineral resources potential

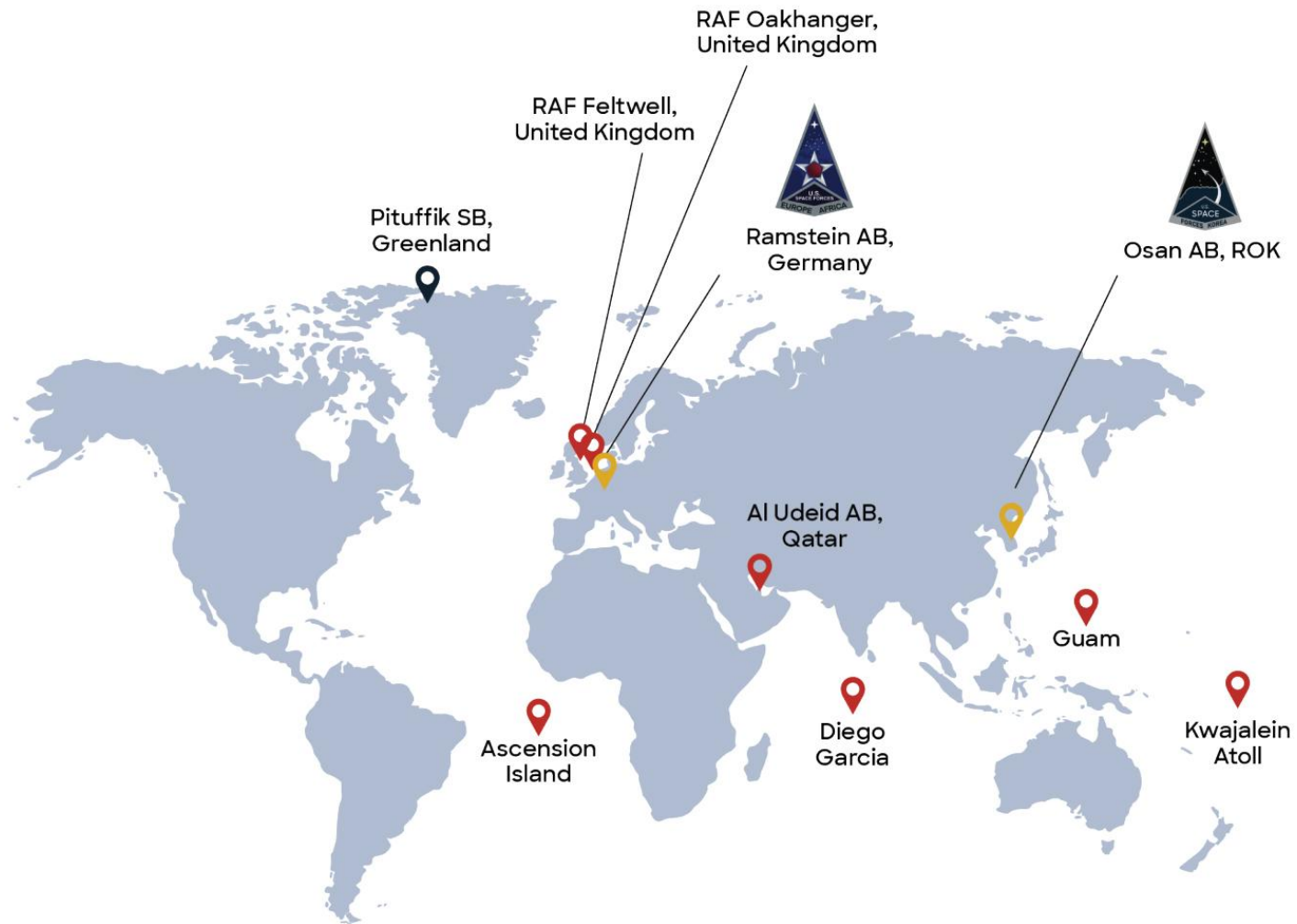
While the country's inland is mostly covered by glaciers, assessments of its coastline show the potential for the extraction of several minerals.

- Copper
- Diamonds
- Gold
- Graphite
- Iron ore
- Nickel
- Rare earths
- Titanium-Vanadium
- Tungsten
- Uranium
- Zinc



Note: Some of Greenland's main potential mineral deposits.
Source: Based on data by the Mineral Resources Authority as of Jan 10 | By Alessandro Parodi





LEGEND

- U.S. Space Force HQ
- U.S. Space Force FLDCOM HQ
- U.S. Space Force Component FLDCOM HQ
- U.S. Space Force Delta HQ / DRU
- U.S. Space Force Base (SFB) / Station (SFS)
- U.S. Space Force Sites



Terminology

Global ice

Glacial hazards

Risk management

Case studies

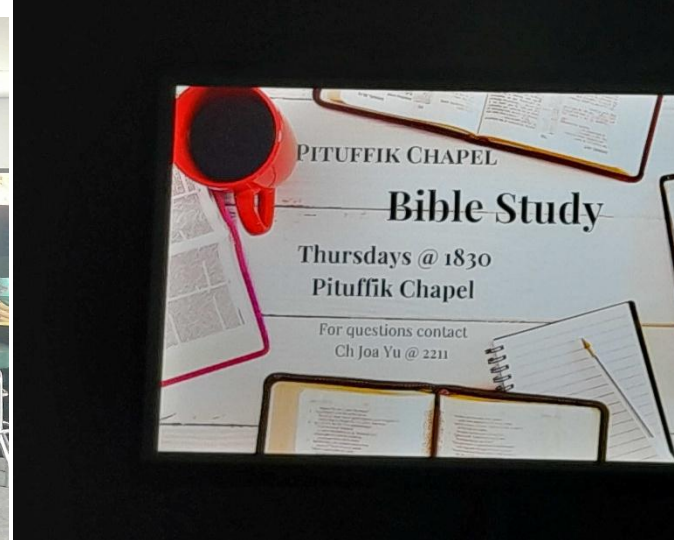
Pituffik Airbase (formerly Thule Airbase)

Built to current extent during the cold war

Serves until today as important satellite downlink and intercontinental rocket early warning site

Displaced local population





Terminology

Global ice

Glacial hazards

Risk management

Case studies



Terminology

Global ice

Glacial hazards

Risk management

Case studies



Terminology

Global ice

Glacial hazards

Risk management

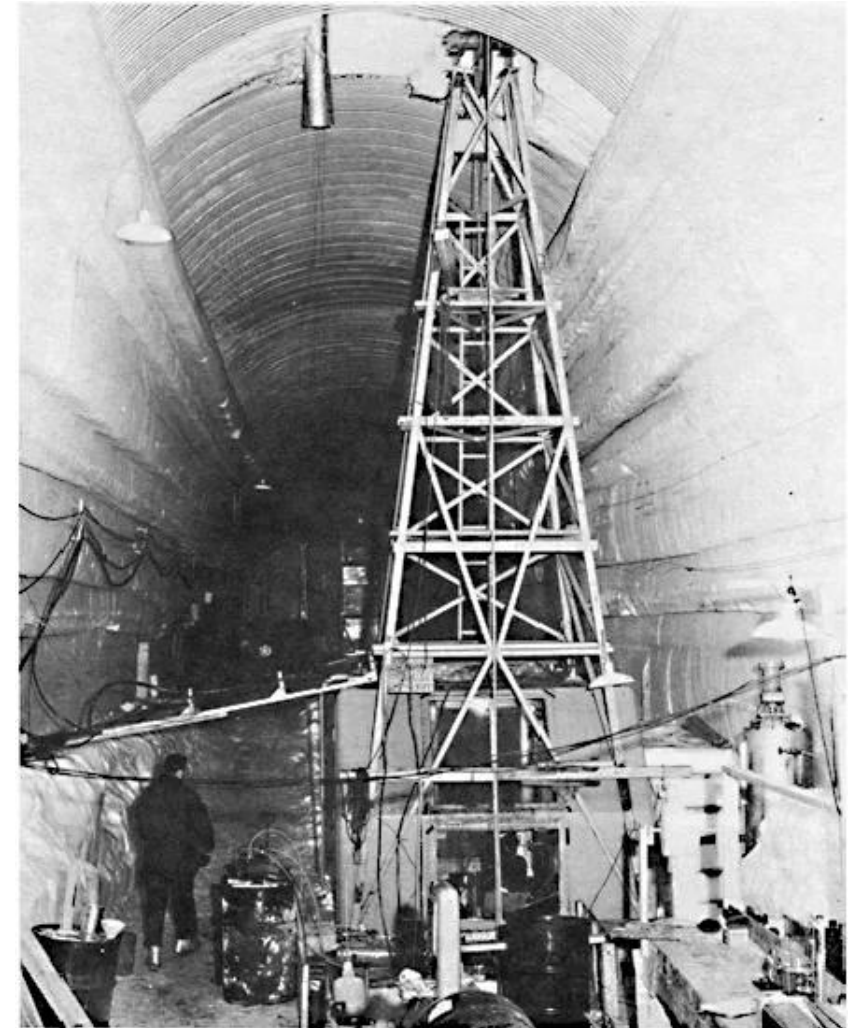
Case studies



Regulations and oversight in an extreme environment?



1968 Thule Air Base B-52 crash



Camp Century (1950s/1960s)

Hazards and Risk Assessment 2026

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fidelsteiner.github.io/#teaching